

## SalesVerse 2.0

# Data Dictionary

Physical data model of the PostgreSQL database: tables, columns, keys, relationships, enumerations and database objects

IIFT Agent/Banca Portal & Back-office for Insurans Islam Family Takaful Sendirian Berhad



### PREPARED FOR

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|                    |                                     |
|--------------------|-------------------------------------|
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## Document Control

This data dictionary describes every table and column of the SalesVerse 2.0 database as created by the versioned migrations in the source code. It is generated from the Prisma schema and the migration SQL, so it always matches the deployed database. A spreadsheet version with the same content is provided for filtering and searching.

### Version history

| Version | Date           | Author        | Description                     |
|---------|----------------|---------------|---------------------------------|
| 0.9     | September 2026 | iorta TechNXT | Internal technical review       |
| 1.0     | October 2026   | iorta TechNXT | Issued with the tender proposal |

### Review and approval

| Role                                            | Name                | Signature / date |
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| Accepted by (IIFT IT)                           | [Name, designation] |                  |

### Document pack

| Reference    | Document                                       | RFP deliverable        | Status                          |
|--------------|------------------------------------------------|------------------------|---------------------------------|
| IIFT-SV2-SAD | SalesVerse 2.0 Solution Architecture           | DEL-06                 | Issued                          |
| IIFT-SV2-DD  | SalesVerse 2.0 Data Dictionary                 | DEL-07                 | This document                   |
| IIFT-SV2-PSH | SalesVerse 2.0 Production Support Handover     | DEL-26, DEL-27, MNT-28 | Issued                          |
| IIFT-SV2-CSQ | SalesVerse 2.0 Code Standards & Quality Report | DEL-11, NFR-14         | Delivered during implementation |
| IIFT-SV2-SAR | SalesVerse 2.0 Security Assessment Report      | DEL-17, NFR-13         | Delivered during implementation |

### Related material

- SalesVerse-2.0-Data-Dictionary.xlsx: the same content as a workbook, one sheet per table.
- SalesVerse-2.0-Solution-Architecture: data architecture, encryption and retention design.
- Source: apps/api/prisma/schema.prisma and apps/api/prisma/migrations/\*/migration.sql.

### Conventions

- Monetary amounts are in Brunei dollars (B\$). Times are Brunei time (UTC+8) unless marked UTC.
- RFP requirement identifiers (AP, BO, INT, COM, NFR, DEL, MNT) refer to IIFT's Request for Proposal. COM-01 to COM-10 in this pack are the shared platform requirements of RFP section 4.4.
- Items marked **delivered during implementation** are designed here but not yet present in the working application; everything else describes the application as it runs today.
- Text in [square brackets] is to be completed with IIFT during mobilisation.

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# 1. Introduction

## 1.1 Scope

This dictionary covers the complete SalesVerse 2.0 database as defined by migration 20261002133922\_initial\_schema: 37 tables, 434 columns, 36 enumerated types, 27 foreign keys, 109 indexes including primary keys, 9 sequences, 4 check constraints and one trigger. Schema changes are made only through new migrations, and this document is regenerated with each release that contains one.

## 1.2 How the dictionary is produced

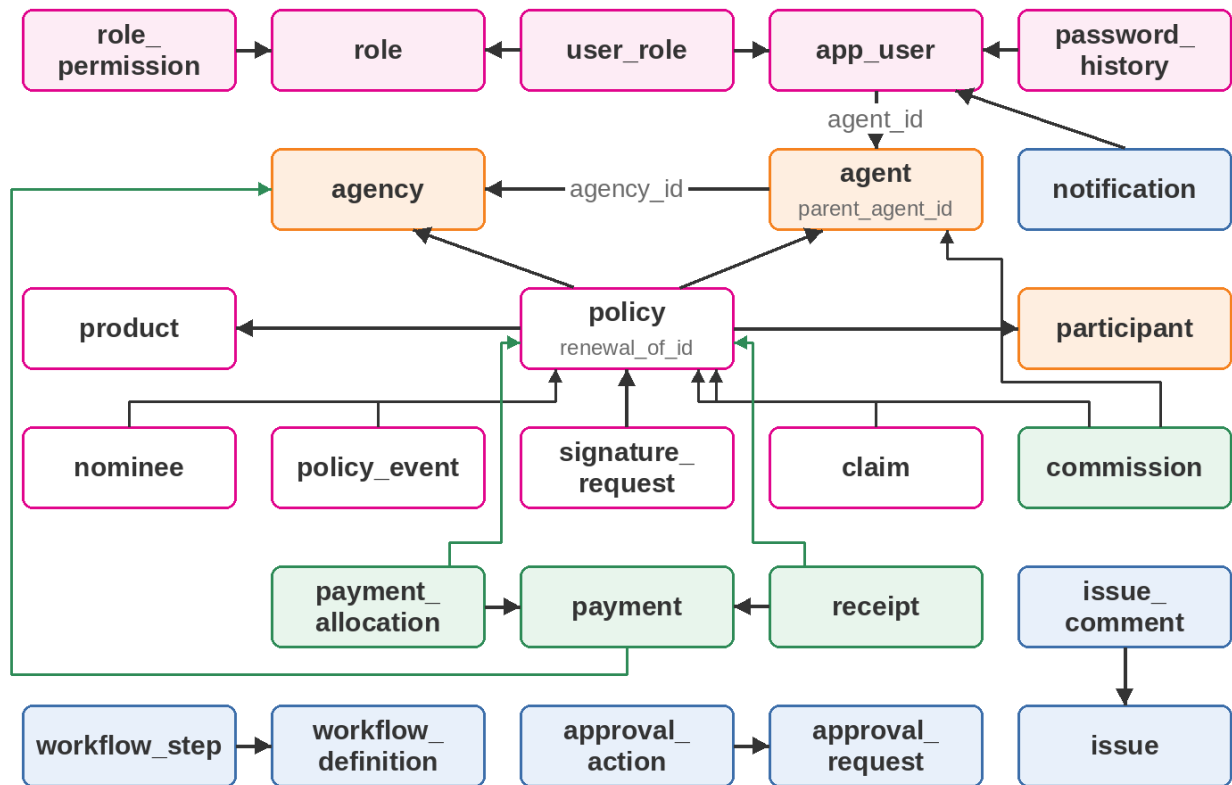
A generator in docs/technical/build reads apps/api/prisma/schema.prisma for tables, columns, keys and relations, and the migration SQL for index names, ON DELETE rules, check constraints, sequences and the audit trigger. It compares every column type and nullability derived from the Prisma schema with the CREATE TABLE statement in the migrations and stops if they differ. Business descriptions are maintained next to the generator, and the build also stops if a table, column, enumeration value, sequence or check constraint has no description.

## 1.3 Conventions

| Item                      | Convention                                                                                                                                                                                                          |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Names</b>              | Tables and columns use snake_case in PostgreSQL (from @@map and @map). The application uses camelCase model and field names; both are listed.                                                                       |
| <b>Primary keys</b>       | UUID values generated by the application, except audit_log (BIGSERIAL, to give a total order of events), config_parameter (natural key), user_session (session id) and the two link tables (composite keys).        |
| <b>Timestamps</b>         | TIMESTAMP(3) without time zone, written in UTC by the application. created_at defaults to CURRENT_TIMESTAMP; updated_at is set by the application on every update.                                                  |
| <b>Business dates</b>     | DATE columns (start_date, payment_date, business_date ...) hold calendar dates in Brunei time (UTC+8).                                                                                                              |
| <b>Money</b>              | DECIMAL(14,2) in Brunei dollars; rates DECIMAL(6,4); percentages DECIMAL(5,2). Amounts are never stored as floating point.                                                                                          |
| <b>Optimistic locking</b> | Tables edited by several users (agent, participant, policy) carry a version column that is checked and incremented on update.                                                                                       |
| <b>Keys column</b>        | PK primary key; UQ unique index; IDX non-unique index; (n/m) is the column's position in a composite index.                                                                                                         |
| <b>JSON</b>               | JSONB is used for configuration and variable structures (product config, questionnaire answers, audit before/after values, outbox payloads). Each JSON column is validated by the application before it is written. |

Table 1: Dictionary conventions

## 1.4 Model overview



### Tables without foreign keys (polymorphic owner, logs or configuration)

document, audit\_log, outbox\_message, integration\_log, reconciliation\_run, eod\_run, report\_schedule, config\_parameter, code\_item, aml\_screening, aml\_watchlist\_entry, user\_session

Arrow: child table points to its parent (foreign key). Grey text: self-reference.

Figure 1: Entity overview: foreign-key relationships

| Table                            | Cols | Description                                                                                                                                                               |
|----------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identity and access</b>       |      |                                                                                                                                                                           |
| app_user                         | 17   | Every person who can sign in: agents and bank officers (portal) and IIFT staff (back-office). Holds credentials, lockout state and the optional link to an agent profile. |
| password_history                 | 4    | Hashes of previous passwords, used to stop reuse of the last N passwords (security.password.history_count).                                                               |
| role                             | 8    | Configurable roles (BO-04). Each role belongs to one audience, portal or back-office, and is built from permissions in role_permission.                                   |
| role_permission                  | 2    | Permissions granted to a role. Permission codes are defined in the application (src/common/security/permissions.ts) because each one guards code.                         |
| user_role                        | 2    | Roles assigned to a user. A user's effective permissions are the union of the permissions of roles that match the user's audience.                                        |
| user_session                     | 3    | Server-side HTTP sessions written by connect-pg-simple. The browser holds only the session id in an HttpOnly cookie; deleting a row ends that session immediately.        |
| <b>Agency / Banca and agents</b> |      |                                                                                                                                                                           |
| agency                           | 14   | Distribution partners: agencies (agency channel) and banks (banca channel). Carries the automatic issuance block applied when contributions are overdue (AP-42).          |
| agent                            | 25   | Agents, sub-agents and bank officers with their hierarchy, licence, status, AML status and authority limit. Identification numbers are encrypted.                         |

| Table                                    | Cols | Description                                                                                                                                                                             |
|------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Participants and AML/KYC</b>          |      |                                                                                                                                                                                         |
| <b>participant</b>                       | 26   | Shared participant (certificate holder) master record, individual or corporate. One record per identification number across all agencies (AP-11). Identification numbers are encrypted. |
| <b>aml_screening</b>                     | 12   | One row per AML/KYC screening of an agent or participant, with match details and the Compliance review decision (BO-13 to BO-15).                                                       |
| <b>aml_watchlist_entry</b>               | 9    | Watch-list entries (sanctions, PEP, internal lists) loaded by Compliance and used by the built-in screening.                                                                            |
| <b>Products, quotations and policies</b> |      |                                                                                                                                                                                         |
| <b>product</b>                           | 14   | Takaful products offered through the portal (Appendix 3) with their rating configuration, required documents and questionnaire, all maintainable without code changes (BO-33).          |
| <b>policy</b>                            | 33   | Quotations and policies. A row starts as a DRAFT quotation and keeps the same id through approval, issuance, renewal, endorsement, cancellation and expiry.                             |
| <b>nominee</b>                           | 7    | Nominees, beneficiaries and executors named on a policy.                                                                                                                                |
| <b>policy_event</b>                      | 9    | Business history of a policy: every status change and significant action with actor and remarks (AP-31).                                                                                |
| <b>signature_request</b>                 | 11   | One-time e-signature links e-mailed to participants (AP-62). Only a hash of the token is stored.                                                                                        |
| <b>Billing</b>                           |      |                                                                                                                                                                                         |
| <b>payment</b>                           | 15   | Contribution payments submitted by agencies and banks, with proof of payment, awaiting or after Finance verification (AP-38 to AP-40).                                                  |
| <b>payment_allocation</b>                | 4    | How a payment is split across the policies it pays for (bulk payment, AP-38).                                                                                                           |
| <b>receipt</b>                           | 7    | Official e-Receipts issued per policy when a payment is verified (AP-41).                                                                                                               |
| <b>commission</b>                        | 11   | Commission or referral fee accrued per agent and policy at issuance, and marked paid when Finance confirms payment (INT-05).                                                            |
| <b>Claims</b>                            |      |                                                                                                                                                                                         |
| <b>claim</b>                             | 14   | Claim notifications submitted through the portal (AP-43). Assessment happens in the core system.                                                                                        |
| <b>Documents</b>                         |      |                                                                                                                                                                                         |
| <b>document</b>                          | 17   | Metadata of every uploaded or system-generated document. File content is held encrypted in the document store and addressed by storage_key.                                             |
| <b>Workflow</b>                          |      |                                                                                                                                                                                         |
| <b>workflow_definition</b>               | 6    | Approval workflow per transaction type (COM-04, BO-16). Inactive definitions approve requests automatically.                                                                            |
| <b>workflow_step</b>                     | 6    | Levels of a workflow: permission required to approve the level and optional amount threshold above which the level applies (authority limits).                                          |
| <b>approval_request</b>                  | 18   | Maker-checker requests raised for a transaction type, with the proposed change as payload and the approval levels resolved at submission.                                               |
| <b>approval_action</b>                   | 8    | Every action on an approval request: submit, approve, reject or withdraw, with actor and remarks (AP-61).                                                                               |
| <b>Issue management</b>                  |      |                                                                                                                                                                                         |
| <b>issue</b>                             | 22   | Issues reported from the portal or back-office with priority, assignment and SLA timestamps (AP-55, BO-29 to BO-31).                                                                    |
| <b>issue_comment</b>                     | 7    | Comments on an issue. Internal comments are visible only to back-office staff.                                                                                                          |
| <b>Notifications</b>                     |      |                                                                                                                                                                                         |
| <b>notification</b>                      | 16   | In-app, e-mail and SMS notifications. E-mail and SMS rows are delivered asynchronously by the notification dispatcher with retry.                                                       |
| <b>Audit trail</b>                       |      |                                                                                                                                                                                         |
| <b>audit_log</b>                         | 12   | Append-only audit trail of security and business events with before and after values (BO-26 to BO-28, COM-03). A trigger rejects UPDATE and DELETE.                                     |
| <b>Configuration and master data</b>     |      |                                                                                                                                                                                         |

| Table                     | Cols | Description                                                                                                                                                                           |
|---------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>config_parameter</b>   | 9    | Business parameters administrators can change without a release: password policy, session timeouts, grace period, SLA hours and similar (BO-33, COM-09).                              |
| <b>code_item</b>          | 6    | Master data code lists (document types, relationships, nationalities, districts, banks, occupations, claim types, issue categories and others) maintained in the back-office (BO-32). |
| <b>Integration</b>        |      |                                                                                                                                                                                       |
| <b>outbox_message</b>     | 13   | Transactional outbox for messages to the core and financial systems. Written in the same transaction as the business change and delivered asynchronously with retry.                  |
| <b>integration_log</b>    | 12   | One row per integration call, outbound or inbound, with outcome and duration (INT-11, INT-14).                                                                                        |
| <b>reconciliation_run</b> | 10   | Results of daily reconciliation between receipts issued by the portal and receipts posted by the financial system (INT-15).                                                           |
| <b>eod_run</b>            | 13   | End-of-day runs per business date with totals and links to the EOD report and FIN interface file.                                                                                     |
| <b>report_schedule</b>    | 12   | Scheduled reports e-mailed to recipients daily, weekly or monthly (BO-25).                                                                                                            |

Table 2: Tables by area

## 2. Tables

Each table lists its columns in creation order. Foreign key shows the referenced table and column and the ON DELETE rule. Protection marks encrypted, blind-indexed or hashed values.

### 2.1 app\_user

Every person who can sign in: agents and bank officers (portal) and IIFT staff (back-office). Holds credentials, lockout state and the optional link to an agent profile. Model User; area Identity and access; primary key id.

| Column               | Data type    | Null | Default                              | Keys / foreign key                     | Description                                                                                                               |
|----------------------|--------------|------|--------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| id                   | UUID         | No   | UUID v4 generated by the application | PK                                     | Surrogate primary key.                                                                                                    |
| username             | VARCHAR(64)  | No   |                                      | UQ                                     | Sign-in name, stored in lower case and unique across all users.                                                           |
| email                | VARCHAR(254) | No   |                                      |                                        | E-mail address for notifications and password reset communication.                                                        |
| full_name            | VARCHAR(150) | No   |                                      |                                        | Name shown on screens, in approval history and in the audit trail.                                                        |
| mobile               | VARCHAR(20)  | Yes  |                                      |                                        | Mobile number for SMS notifications, in international format.                                                             |
| user_type            | UserType     | No   |                                      | IDX (1/2)                              | AGENT, BANCA or STAFF. STAFF users belong to the back-office audience; the others to the portal.                          |
| status               | UserStatus   | No   | 'ACTIVE'                             | IDX (2/2)                              | ACTIVE, LOCKED (temporarily, after failed sign-ins) or DISABLED (by an administrator or core-system status sync).         |
| auth_source          | AuthSource   | No   | 'LOCAL'                              |                                        | LOCAL for passwords held by the application; DIRECTORY for back-office users authenticated against Active Directory/LDAP. |
| password_hash        | TEXT         | Yes  |                                      |                                        | Argon2id hash of the password (memory 19 MiB, 2 iterations). Null for directory accounts. <b>One-way hash (Argon2id).</b> |
| password_changed_at  | TIMESTAMP(3) | Yes  |                                      |                                        | When the password was last changed; drives password expiry.                                                               |
| must_change_password | BOOLEAN      | No   | true                                 |                                        | True when the user must change the password at next sign-in (new account or administrator reset).                         |
| failed_login_count   | INTEGER      | No   | 0                                    |                                        | Consecutive failed sign-in attempts since the last success or lock.                                                       |
| locked_until         | TIMESTAMP(3) | Yes  |                                      |                                        | End of the current lockout period; null when not locked.                                                                  |
| last_login_at        | TIMESTAMP(3) | Yes  |                                      |                                        | Time of the last successful sign-in.                                                                                      |
| agent_id             | UUID         | Yes  |                                      | UQ<br>FK agent(id); on delete set null | Agent profile of a portal user; null for back-office staff. One user per agent.                                           |
| created_at           | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                    |                                        | Date and time the row was created (UTC).                                                                                  |

| Column     | Data type    | Null | Default                                | Keys / foreign key | Description                                                                  |
|------------|--------------|------|----------------------------------------|--------------------|------------------------------------------------------------------------------|
| updated_at | TIMESTAMP(3) | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application. |

## 2.2 password\_history

Hashes of previous passwords, used to stop reuse of the last N passwords (security.password.history\_count). Model PasswordHistory; area Identity and access; primary key id.

| Column        | Data type    | Null | Default                              | Keys / foreign key                              | Description                                                           |
|---------------|--------------|------|--------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------|
| id            | UUID         | No   | UUID v4 generated by the application | PK                                              | Surrogate primary key.                                                |
| user_id       | UUID         | No   |                                      | IDX (1/2)<br>FK app_user(id); on delete cascade | User the historical password belonged to.                             |
| password_hash | TEXT         | No   |                                      |                                                 | Argon2id hash of a previous password. <b>One-way hash (Argon2id).</b> |
| created_at    | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                    | IDX (2/2)                                       | Date and time the row was created (UTC).                              |

## 2.3 role

Configurable roles (BO-04). Each role belongs to one audience, portal or back-office, and is built from permissions in role\_permission. Model Role; area Identity and access; primary key id.

| Column      | Data type    | Null | Default                                | Keys / foreign key | Description                                                                             |
|-------------|--------------|------|----------------------------------------|--------------------|-----------------------------------------------------------------------------------------|
| id          | UUID         | No   | UUID v4 generated by the application   | PK                 | Surrogate primary key.                                                                  |
| code        | VARCHAR(50)  | No   |                                        | UQ                 | Stable role code used by seed data and integrations, e.g. OPERATIONS_SUPERVISOR.        |
| name        | VARCHAR(100) | No   |                                        |                    | Role name shown to administrators.                                                      |
| description | VARCHAR(500) | Yes  |                                        |                    | What the role is for.                                                                   |
| audience    | Audience     | No   |                                        |                    | PORTAL or BACKOFFICE. Only permissions of the same audience can be granted to the role. |
| is_system   | BOOLEAN      | No   | false                                  |                    | True for roles created by the seed script. System roles can be edited but not deleted.  |
| created_at  | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                      |                    | Date and time the row was created (UTC).                                                |
| updated_at  | TIMESTAMP(3) | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application.            |

## 2.4 role\_permission

Permissions granted to a role. Permission codes are defined in the application (src/common/security/permissions.ts) because each one guards code. Model RolePermission; area Identity and access; primary key role\_id, permission.

| Column     | Data type   | Null | Default | Keys / foreign key                   | Description                                                                |
|------------|-------------|------|---------|--------------------------------------|----------------------------------------------------------------------------|
| role_id    | UUID        | No   |         | PK<br>FK role(id); on delete cascade | Role receiving the permission.                                             |
| permission | VARCHAR(80) | No   |         | PK                                   | Permission code, e.g. bo.approve.policies, from the application catalogue. |

## 2.5 user\_role

Roles assigned to a user. A user's effective permissions are the union of the permissions of roles that match the user's audience. Model UserRole; area Identity and access; primary key user\_id, role\_id.

| Column  | Data type | Null | Default | Keys / foreign key                       | Description                |
|---------|-----------|------|---------|------------------------------------------|----------------------------|
| user_id | UUID      | No   |         | PK<br>FK app_user(id); on delete cascade | User holding the role.     |
| role_id | UUID      | No   |         | PK<br>FK role(id); on delete cascade     | Role assigned to the user. |

## 2.6 user\_session

Server-side HTTP sessions written by connect-pg-simple. The browser holds only the session id in an HttpOnly cookie; deleting a row ends that session immediately. Model UserSession; area Identity and access; primary key sid.

| Column | Data type    | Null | Default | Keys / foreign key | Description                                                                                                                                                                          |
|--------|--------------|------|---------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sid    | VARCHAR      | No   |         | PK                 | Session identifier carried in the session cookie. Regenerated at sign-in.                                                                                                            |
| sess   | JSON         | No   |         |                    | Session data: signed-in user snapshot (identity, audience, permissions, agency link), CSRF token and authentication time. <b>Holds the CSRF token; never exposed to the browser.</b> |
| expire | TIMESTAMP(6) | No   |         | IDX                | When the session expires if idle. Expired rows are pruned every 15 minutes.                                                                                                          |

## 2.7 agency

Distribution partners: agencies (agency channel) and banks (banca channel). Carries the automatic issuance block applied when contributions are overdue (AP-42). Model Agency; area Agency / Banca and agents; primary key id.

| Column                | Data type    | Null | Default                                | Keys / foreign key | Description                                                                                            |
|-----------------------|--------------|------|----------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------|
| id                    | UUID         | No   | UUID v4 generated by the application   | PK                 | Surrogate primary key.                                                                                 |
| code                  | VARCHAR(20)  | No   |                                        | UQ                 | Unique agency or bank code used on reports and the FIN interface file.                                 |
| name                  | VARCHAR(150) | No   |                                        |                    | Registered name of the agency or bank.                                                                 |
| channel               | Channel      | No   |                                        |                    | AGENCY or BANCA distribution channel.                                                                  |
| registration_no       | VARCHAR(50)  | Yes  |                                        |                    | Business registration number of the agency or bank.                                                    |
| email                 | VARCHAR(254) | Yes  |                                        |                    | Main contact e-mail address.                                                                           |
| phone                 | VARCHAR(20)  | Yes  |                                        |                    | Main contact telephone number.                                                                         |
| address               | VARCHAR(300) | Yes  |                                        |                    | Business address.                                                                                      |
| status                | AgencyStatus | No   | 'ACTIVE'                               |                    | ACTIVE, INACTIVE or SUSPENDED. Users of a non-active agency cannot sign in.                            |
| issuance_blocked      | BOOLEAN      | No   | false                                  |                    | True while new business is blocked because an issued policy is unpaid beyond the grace period (AP-42). |
| issuance_blocked_at   | TIMESTAMP(3) | Yes  |                                        |                    | When the current block was applied.                                                                    |
| issuance_block_reason | VARCHAR(300) | Yes  |                                        |                    | Reason shown to the agency, including the number of overdue policies and the amount.                   |
| created_at            | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                      |                    | Date and time the row was created (UTC).                                                               |
| updated_at            | TIMESTAMP(3) | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application.                           |

## 2.8 agent

Agents, sub-agents and bank officers with their hierarchy, licence, status, AML status and authority limit. Identification numbers are encrypted. Model Agent; area Agency / Banca and agents; primary key id.

| Column     | Data type   | Null | Default                              | Keys / foreign key | Description                                                                                                 |
|------------|-------------|------|--------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------|
| id         | UUID        | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                                      |
| agent_code | VARCHAR(20) | No   |                                      | UQ                 | Unique agent code, AG-nnnnnn for agency agents and BK-nnnnnn for bank officers, issued from seq_agent_code. |

| Column          | Data type     | Null | Default                                | Keys / foreign key                             | Description                                                                                                                                                                                              |
|-----------------|---------------|------|----------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agency_id       | UUID          | No   |                                        | IDX (1/2)<br>FK agency(id); on delete restrict | Agency or bank the agent belongs to.                                                                                                                                                                     |
| agent_type      | AgentType     | No   |                                        |                                                | MAIN_AGENT, SUB_AGENT or BANKER.                                                                                                                                                                         |
| parent_agent_id | UUID          | Yes  |                                        | IDX<br>FK agent(id); on delete set null        | Main agent a sub-agent reports to; null for main agents and bank officers without a supervisor.                                                                                                          |
| full_name       | VARCHAR(150)  | No   |                                        |                                                | Agent's full name as on the identity document.                                                                                                                                                           |
| id_type         | IdType        | No   |                                        | UQ (1/2)                                       | Type of identification: NRIC or PASSPORT.                                                                                                                                                                |
| id_number_enc   | TEXT          | No   |                                        |                                                | Identification number encrypted with AES-256-GCM (format v1:<base64 iv[tag  ciphertext>). Decrypted only for authorised display, and then masked. <b>Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)</b> . |
| id_number_hash  | VARCHAR(64)   | No   |                                        | UQ (2/2)                                       | Blind index: HMAC-SHA256 of the normalised identification number. Allows exact search and the uniqueness check without decrypting. <b>Blind index (HMAC-SHA256, FIELD_HASH_KEY)</b> .                    |
| date_of_birth   | DATE          | Yes  |                                        |                                                | Date of birth.                                                                                                                                                                                           |
| email           | VARCHAR(254)  | No   |                                        |                                                | Agent's e-mail address for notifications.                                                                                                                                                                |
| mobile          | VARCHAR(20)   | No   |                                        |                                                | Agent's mobile number for SMS notifications.                                                                                                                                                             |
| address         | VARCHAR(300)  | Yes  |                                        |                                                | Correspondence address.                                                                                                                                                                                  |
| branch_name     | VARCHAR(100)  | Yes  |                                        |                                                | Bank branch of a bank officer.                                                                                                                                                                           |
| licence_no      | VARCHAR(50)   | Yes  |                                        |                                                | Agent licence or registration number.                                                                                                                                                                    |
| licence_expiry  | DATE          | Yes  |                                        |                                                | Licence expiry date, used for expiry warnings.                                                                                                                                                           |
| status          | AgentStatus   | No   | 'PENDING'                              | IDX (2/2)                                      | Registration status: PENDING approval, ACTIVE, INACTIVE, SUSPENDED, TERMINATED or REJECTED.                                                                                                              |
| status_reason   | VARCHAR(300)  | Yes  |                                        |                                                | Reason for the latest status change, including changes received from the core system.                                                                                                                    |
| aml_status      | AmlStatus     | No   | 'NOT_SCREENED'                         |                                                | Latest AML/KYC outcome: NOT_SCREENED, CLEAR, FLAGGED or REJECTED.                                                                                                                                        |
| authority_limit | DECIMAL(14,2) | Yes  |                                        |                                                | Highest sum covered (B\$) the agent may submit without referral; above it the quotation is referred for approval (AP-60). Null means no personal limit.                                                  |
| activated_at    | TIMESTAMP(3)  | Yes  |                                        |                                                | When the registration was approved and the agent activated.                                                                                                                                              |
| terminated_at   | TIMESTAMP(3)  | Yes  |                                        |                                                | When the agent was terminated.                                                                                                                                                                           |
| created_at      | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP                      |                                                | Date and time the row was created (UTC).                                                                                                                                                                 |
| updated_at      | TIMESTAMP(3)  | No   | Set by the application on every update |                                                | Date and time the row was last changed (UTC), maintained by the application.                                                                                                                             |

| Column  | Data type | Null | Default | Keys / foreign key | Description                                                                                                                                                                   |
|---------|-----------|------|---------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| version | INTEGER   | No   | 0       |                    | Optimistic-lock counter. Incremented on every update; an update that quotes a stale version is rejected with STALE_RECORD so two users cannot overwrite each other's changes. |

## 2.9 participant

Shared participant (certificate holder) master record, individual or corporate. One record per identification number across all agencies (AP-11). Identification numbers are encrypted. Model Participant; area Participants and AML/KYC; primary key id.

| Column           | Data type       | Null | Default                              | Keys / foreign key | Description                                                                                                                                                                                        |
|------------------|-----------------|------|--------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id               | UUID            | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                                                                                                                             |
| participant_no   | VARCHAR(20)     | No   |                                      | UQ                 | Business reference PT/yy/nnnnnn issued from seq_participant_no.                                                                                                                                    |
| type             | ParticipantType | No   |                                      |                    | INDIVIDUAL or CORPORATE.                                                                                                                                                                           |
| full_name        | VARCHAR(150)    | No   |                                      | IDX                | Full name of the individual or registered name of the company.                                                                                                                                     |
| id_type          | IdType          | No   |                                      | UQ (1/2)           | NRIC, PASSPORT or BUSINESS_REG.                                                                                                                                                                    |
| id_number_enc    | TEXT            | No   |                                      |                    | Identification number encrypted with AES-256-GCM; displayed masked. <b>Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY).</b>                                                                          |
| id_number_hash   | VARCHAR(64)     | No   |                                      | UQ (2/2)           | Blind index (HMAC-SHA256) of the normalised identification number; enforces one participant per identification number and supports exact search. <b>Blind index (HMAC-SHA256, FIELD_HASH_KEY).</b> |
| date_of_birth    | DATE            | Yes  |                                      |                    | Date of birth; drives age-next-birthday rating and eligibility.                                                                                                                                    |
| gender           | VARCHAR(10)     | Yes  |                                      |                    | Gender code from master data.                                                                                                                                                                      |
| nationality      | VARCHAR(50)     | Yes  |                                      |                    | Nationality code from master data; used by eligibility rules (e.g. OSA).                                                                                                                           |
| occupation       | VARCHAR(100)    | Yes  |                                      |                    | Occupation code from master data.                                                                                                                                                                  |
| occupation_class | INTEGER         | Yes  |                                      |                    | Occupational class 1 to 4; the Professional plan accepts class I only.                                                                                                                             |
| email            | VARCHAR(254)    | Yes  |                                      |                    | Participant e-mail for documents and e-signature links.                                                                                                                                            |
| mobile           | VARCHAR(20)     | No   |                                      |                    | Participant mobile number.                                                                                                                                                                         |
| address_line1    | VARCHAR(200)    | No   |                                      |                    | First address line.                                                                                                                                                                                |
| address_line2    | VARCHAR(200)    | Yes  |                                      |                    | Second address line.                                                                                                                                                                               |
| postcode         | VARCHAR(10)     | Yes  |                                      |                    | Postcode.                                                                                                                                                                                          |
| district         | VARCHAR(50)     | Yes  |                                      |                    | District code from master data.                                                                                                                                                                    |

| Column               | Data type    | Null | Default                                | Keys / foreign key | Description                                                                                                                                                                   |
|----------------------|--------------|------|----------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| contact_person       | VARCHAR(150) | Yes  |                                        |                    | Contact person for a corporate participant.                                                                                                                                   |
| aml_status           | AmlStatus    | No   | 'NOT_SCREENED'                         |                    | Latest AML/KYC outcome. FLAGGED or REJECTED blocks submission of applications.                                                                                                |
| aml_screened_at      | TIMESTAMP(3) | Yes  |                                        |                    | When the participant was last screened.                                                                                                                                       |
| created_by_agent_id  | UUID         | Yes  |                                        |                    | Agent who registered the participant (no foreign key, kept for history).                                                                                                      |
| created_by_agency_id | UUID         | Yes  |                                        | IDX                | Agency of the registering agent; used for data scoping.                                                                                                                       |
| created_at           | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                      |                    | Date and time the row was created (UTC).                                                                                                                                      |
| updated_at           | TIMESTAMP(3) | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application.                                                                                                  |
| version              | INTEGER      | No   | 0                                      |                    | Optimistic-lock counter. Incremented on every update; an update that quotes a stale version is rejected with STALE_RECORD so two users cannot overwrite each other's changes. |

## 2.10 aml\_screening

One row per AML/KYC screening of an agent or participant, with match details and the Compliance review decision (BO-13 to BO-15). Model AmlScreening; area Participants and AML/KYC; primary key id.

| Column         | Data type     | Null | Default                              | Keys / foreign key | Description                                                                                       |
|----------------|---------------|------|--------------------------------------|--------------------|---------------------------------------------------------------------------------------------------|
| id             | UUID          | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                            |
| subject_type   | VARCHAR(20)   | No   |                                      | IDX (1/2)          | AGENT or PARTICIPANT.                                                                             |
| subject_id     | UUID          | No   |                                      | IDX (2/2)          | Id of the screened agent or participant.                                                          |
| subject_name   | VARCHAR(150)  | No   |                                      |                    | Name screened, kept as at the time of screening.                                                  |
| provider       | VARCHAR(50)   | No   |                                      |                    | Source of the result: WATCHLIST, WATCHLIST+EXTERNAL or WATCHLIST+EXTERNAL_UNAVAILABLE.            |
| score          | INTEGER       | No   |                                      |                    | Highest match score, 0 to 100. At or above aml.match_threshold the case is flagged.               |
| matches        | JSONB         | No   |                                      |                    | Up to ten best matches: list name, matched name, score, reference and reason (NAME or ID_NUMBER). |
| status         | AmlCaseStatus | No   |                                      | IDX                | AUTO_CLEARED, PENDING_REVIEW, CLEARED or CONFIRMED_MATCH.                                         |
| reviewed_by_id | UUID          | Yes  |                                      |                    | Compliance officer who reviewed the case.                                                         |
| reviewed_at    | TIMESTAMP(3)  | Yes  |                                      |                    | When the case was reviewed.                                                                       |

| Column         | Data type     | Null | Default           | Keys / foreign key | Description                                         |
|----------------|---------------|------|-------------------|--------------------|-----------------------------------------------------|
| review_remarks | VARCHAR(1000) | Yes  |                   |                    | Mandatory reviewer remarks explaining the decision. |
| created_at     | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP |                    | Date and time the row was created (UTC).            |

## 2.11 aml\_watchlist\_entry

Watch-list entries (sanctions, PEP, internal lists) loaded by Compliance and used by the built-in screening. Model AmlWatchlistEntry; area Participants and AML/KYC; primary key id.

| Column     | Data type    | Null | Default                              | Keys / foreign key | Description                                                                                                |
|------------|--------------|------|--------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------|
| id         | UUID         | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                                     |
| list_name  | VARCHAR(50)  | No   |                                      |                    | Name of the list the entry came from, e.g. UN, LOCAL, INTERNAL.                                            |
| full_name  | VARCHAR(150) | No   |                                      |                    | Listed name as published.                                                                                  |
| normalised | VARCHAR(150) | No   |                                      | IDX                | Name in upper case without punctuation or honorifics; used for candidate selection and similarity scoring. |
| id_number  | VARCHAR(50)  | Yes  |                                      |                    | Identification number of the listed person, where published. An exact match scores 100.                    |
| country    | VARCHAR(50)  | Yes  |                                      |                    | Country associated with the entry.                                                                         |
| reference  | VARCHAR(100) | Yes  |                                      |                    | Reference of the entry in the source list.                                                                 |
| active     | BOOLEAN      | No   | true                                 |                    | False when the entry was withdrawn or replaced by a newer import of the list.                              |
| created_at | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                    |                    | Date and time the row was created (UTC).                                                                   |

## 2.12 product

Takaful products offered through the portal (Appendix 3) with their rating configuration, required documents and questionnaire, all maintainable without code changes (BO-33). Model Product; area Products, quotations and policies; primary key id.

| Column           | Data type    | Null | Default                              | Keys / foreign key | Description                                                      |
|------------------|--------------|------|--------------------------------------|--------------------|------------------------------------------------------------------|
| id               | UUID         | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                           |
| code             | VARCHAR(20)  | No   |                                      | UQ                 | Product code, e.g. FTP-HP, PRO, KHR. Prefixes the policy number. |
| name             | VARCHAR(150) | No   |                                      |                    | Product name shown in the catalogue and on documents.            |
| line_of_business | VARCHAR(50)  | No   |                                      |                    | Line of business, e.g. Mortgage Takaful or Annual.               |

| Column                  | Data type     | Null | Default                                | Keys / foreign key | Description                                                                                                                                                   |
|-------------------------|---------------|------|----------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| description             | VARCHAR(1000) | No   |                                        |                    | Short description shown in the product catalogue.                                                                                                             |
| rating_engine           | VARCHAR(30)   | No   |                                        |                    | Rating engine that interprets config: FINANCING (decreasing-term financing products) or FIXED_PLAN (plan-based annual products).                              |
| config                  | JSONB         | No   |                                        |                    | Rating configuration: rate tables, plans, loadings, limits, eligibility, wakalah fee and commission rates. Validated by the rating engine before it is saved. |
| required_documents      | JSONB         | No   |                                        |                    | Documents required for the product: type, label and whether mandatory before submission.                                                                      |
| questionnaire           | JSONB         | No   |                                        |                    | Health or risk questions; a 'yes' to a question marked referIfYes refers the application for approval.                                                        |
| payment_before_issuance | BOOLEAN       | No   | true                                   |                    | True for pay-first products; false for issue-then-pay products, which must be paid within the grace period.                                                   |
| allow_renewal           | BOOLEAN       | No   | true                                   |                    | Whether policies of the product can be renewed from the portal.                                                                                               |
| active                  | BOOLEAN       | No   | true                                   |                    | Inactive products are hidden from the catalogue and cannot be quoted.                                                                                         |
| sort_order              | INTEGER       | No   | 0                                      |                    | Display order in the catalogue.                                                                                                                               |
| updated_at              | TIMESTAMP(3)  | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application.                                                                                  |

## 2.13 policy

Quotations and policies. A row starts as a DRAFT quotation and keeps the same id through approval, issuance, renewal, endorsement, cancellation and expiry. Model Policy; area Products, quotations and policies; primary key id.

| Column         | Data type   | Null | Default                              | Keys / foreign key                             | Description                                                                                         |
|----------------|-------------|------|--------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| id             | UUID        | No   | UUID v4 generated by the application | PK                                             | Surrogate primary key.                                                                              |
| quotation_no   | VARCHAR(20) | No   |                                      | UQ                                             | Quotation reference QT/yy/nnnnnn issued from seq_quotation_no when the draft is created.            |
| policy_no      | VARCHAR(20) | Yes  |                                      | UQ                                             | Policy number <product code>/yy/nnnnnn issued from seq_policy_no at issuance; null before issuance. |
| product_id     | UUID        | No   |                                      | FK product(id); on delete restrict             | Product quoted.                                                                                     |
| participant_id | UUID        | No   |                                      | IDX<br>FK participant(id); on delete restrict  | Participant (certificate holder).                                                                   |
| agent_id       | UUID        | No   |                                      | IDX (1/2)<br>FK agent(id); on delete restrict  | Agent or bank officer who owns the business.                                                        |
| agency_id      | UUID        | No   |                                      | IDX (1/2)<br>FK agency(id); on delete restrict | Agency or bank of the agent; used for data scoping and the grace-period rule.                       |

| Column                 | Data type           | Null | Default                                | Keys / foreign key                | Description                                                                                                                                  |
|------------------------|---------------------|------|----------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| status                 | PolicyStatus        | No   | 'DRAFT'                                | IDX (2/2), IDX (1/3)              | Lifecycle status: DRAFT, PENDING_APPROVAL, PENDING_PAYMENT, ACTIVE, REJECTED, EXPIRED or CANCELLED.                                          |
| payment_status         | PolicyPaymentStatus | No   | 'UNPAID'                               | IDX (2/3)                         | UNPAID, PENDING_VERIFICATION (payment submitted) or PAID.                                                                                    |
| plan_code              | VARCHAR(20)         | Yes  |                                        |                                   | Selected plan for plan-based products, e.g. A, B, C.                                                                                         |
| coverage_type          | VARCHAR(20)         | Yes  |                                        |                                   | Coverage option, e.g. INDIVIDUAL or WIDER for Khairat.                                                                                       |
| sum_covered            | DECIMAL(14,2)       | No   |                                        |                                   | Sum covered in B\$.                                                                                                                          |
| term_months            | INTEGER             | No   |                                        |                                   | Cover term in months.                                                                                                                        |
| start_date             | DATE                | Yes  |                                        |                                   | Cover start date.                                                                                                                            |
| end_date               | DATE                | Yes  |                                        |                                   | Cover end date; the expiry job sets status EXPIRED after this date.                                                                          |
| contribution           | DECIMAL(14,2)       | No   |                                        |                                   | Total contribution payable in B\$, including any additional cover.                                                                           |
| contribution_breakdown | JSONB               | No   |                                        |                                   | Contribution lines, tabarru' and wakalah fee split and commission rate, as rated.                                                            |
| risk_details           | JSONB               | No   |                                        |                                   | Product-specific rating inputs, e.g. financing amount, tenure and profit rate, employer details, student institution.                        |
| questionnaire          | JSONB               | Yes  |                                        |                                   | Answers to the product questionnaire.                                                                                                        |
| referral_reasons       | TEXT[]              | Yes  | Empty array                            |                                   | Reasons the application was referred for approval (authority limit, high-risk limit, questionnaire, quality check). Empty when not referred. |
| outstanding_amount     | DECIMAL(14,2)       | No   | 0                                      |                                   | Contribution not yet paid in B\$.                                                                                                            |
| payment_due_date       | DATE                | Yes  |                                        | IDX (3/3)                         | Last day to submit payment for issue-then-pay products (issuance date plus grace days).                                                      |
| submitted_at           | TIMESTAMP(3)        | Yes  |                                        |                                   | When the quotation was last submitted.                                                                                                       |
| approved_at            | TIMESTAMP(3)        | Yes  |                                        |                                   | When the referral was approved.                                                                                                              |
| issued_at              | TIMESTAMP(3)        | Yes  |                                        | IDX                               | When the policy was issued; drives the EOD selection.                                                                                        |
| rejected_reason        | VARCHAR(1000)       | Yes  |                                        |                                   | Reason given by the checker when the application was rejected (AP-51).                                                                       |
| cancelled_at           | TIMESTAMP(3)        | Yes  |                                        |                                   | When the policy was cancelled.                                                                                                               |
| cancellation_reason    | VARCHAR(1000)       | Yes  |                                        |                                   | Approved cancellation reason.                                                                                                                |
| renewal_of_id          | UUID                | Yes  |                                        | FK policy(id); on delete set null | Policy this one renews; null for new business.                                                                                               |
| created_by_id          | UUID                | No   |                                        |                                   | User who created the quotation.                                                                                                              |
| created_at             | TIMESTAMP(3)        | No   | CURRENT_TIMESTAMP                      |                                   | Date and time the row was created (UTC).                                                                                                     |
| updated_at             | TIMESTAMP(3)        | No   | Set by the application on every update |                                   | Date and time the row was last changed (UTC), maintained by the application.                                                                 |

| Column  | Data type | Null | Default | Keys / foreign key | Description                                                                                                                                                                   |
|---------|-----------|------|---------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| version | INTEGER   | No   | 0       |                    | Optimistic-lock counter. Incremented on every update; an update that quotes a stale version is rejected with STALE_RECORD so two users cannot overwrite each other's changes. |

## 2.14 nominee

Nominees, beneficiaries and executors named on a policy. Model Nominee; area Products, quotations and policies; primary key id.

| Column        | Data type    | Null | Default                              | Keys / foreign key                      | Description                                                                                                                    |
|---------------|--------------|------|--------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| id            | UUID         | No   | UUID v4 generated by the application | PK                                      | Surrogate primary key.                                                                                                         |
| policy_id     | UUID         | No   |                                      | IDX<br>FK policy(id); on delete cascade | Policy the nominee belongs to.                                                                                                 |
| full_name     | VARCHAR(150) | No   |                                      |                                         | Nominee's full name.                                                                                                           |
| id_number_enc | TEXT         | Yes  |                                      |                                         | Nominee's identification number encrypted with AES-256-GCM; optional.<br><b>Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY).</b> |
| relationship  | VARCHAR(50)  | No   |                                      |                                         | Relationship to the participant, from master data.                                                                             |
| role          | NomineeRole  | No   | 'NOMINEE'                            |                                         | NOMINEE, BENEFICIARY or EXECUTOR.                                                                                              |
| share_percent | DECIMAL(5,2) | No   |                                      |                                         | Share of the benefit in per cent, greater than 0 and at most 100; shares on a policy total 100.                                |

## 2.15 policy\_event

Business history of a policy: every status change and significant action with actor and remarks (AP-31). Model PolicyEvent; area Products, quotations and policies; primary key id.

| Column      | Data type     | Null | Default                              | Keys / foreign key                            | Description                                                                                                    |
|-------------|---------------|------|--------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| id          | UUID          | No   | UUID v4 generated by the application | PK                                            | Surrogate primary key.                                                                                         |
| policy_id   | UUID          | No   |                                      | IDX (1/2)<br>FK policy(id); on delete cascade | Policy the event belongs to.                                                                                   |
| action      | VARCHAR(50)   | No   |                                      |                                               | Event code, e.g. QUOTATION_CREATED, SUBMITTED, REFERRED, ISSUED, PARTICIPANT_SIGNED, PAYMENT_APPLIED, EXPIRED. |
| from_status | PolicyStatus  | Yes  |                                      |                                               | Status before the event, when the event changed status.                                                        |
| to_status   | PolicyStatus  | Yes  |                                      |                                               | Status after the event.                                                                                        |
| remarks     | VARCHAR(1000) | Yes  |                                      |                                               | Free-text detail, e.g. referral reasons or rejection remarks.                                                  |

| Column     | Data type    | Null | Default           | Keys / foreign key | Description                                                           |
|------------|--------------|------|-------------------|--------------------|-----------------------------------------------------------------------|
| actor_id   | UUID         | Yes  |                   |                    | User who performed the action; null for system jobs and participants. |
| actor_name | VARCHAR(150) | Yes  |                   |                    | Name of the actor at the time: user name, participant name or System. |
| created_at | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP | IDX (2/2)          | Date and time the row was created (UTC).                              |

## 2.16 signature\_request

One-time e-signature links e-mailed to participants (AP-62). Only a hash of the token is stored. Model SignatureRequest; area Products, quotations and policies; primary key id.

| Column          | Data type    | Null | Default                              | Keys / foreign key                      | Description                                                                                                                               |
|-----------------|--------------|------|--------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| id              | UUID         | No   | UUID v4 generated by the application | PK                                      | Surrogate primary key.                                                                                                                    |
| policy_id       | UUID         | No   |                                      | IDX<br>FK policy(id); on delete cascade | Quotation to be signed.                                                                                                                   |
| token_hash      | VARCHAR(64)  | No   |                                      | UQ                                      | SHA-256 of the one-time token in the e-mailed link. The token itself is never stored.<br><b>One-way hash (SHA-256 of one-time token).</b> |
| recipient_name  | VARCHAR(150) | No   |                                      |                                         | Participant name the signer must type to confirm identity.                                                                                |
| recipient_email | VARCHAR(254) | No   |                                      |                                         | Address the link was sent to.                                                                                                             |
| expires_at      | TIMESTAMP(3) | No   |                                      |                                         | Link expiry (policy.esign_link_hours after sending).                                                                                      |
| signed_at       | TIMESTAMP(3) | Yes  |                                      |                                         | When the participant signed; a non-null value makes the link unusable.                                                                    |
| signer_ip       | VARCHAR(64)  | Yes  |                                      |                                         | IP address the signature was made from.                                                                                                   |
| document_id     | UUID         | Yes  |                                      |                                         | Signature image stored as a document.                                                                                                     |
| created_by_id   | UUID         | No   |                                      |                                         | Agent user who sent the link.                                                                                                             |
| created_at      | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP                    |                                         | Date and time the row was created (UTC).                                                                                                  |

## 2.17 payment

Contribution payments submitted by agencies and banks, with proof of payment, awaiting or after Finance verification (AP-38 to AP-40). Model Payment; area Billing; primary key id.

| Column | Data type | Null | Default                              | Keys / foreign key | Description            |
|--------|-----------|------|--------------------------------------|--------------------|------------------------|
| id     | UUID      | No   | UUID v4 generated by the application | PK                 | Surrogate primary key. |

| Column           | Data type     | Null | Default                | Keys / foreign key                             | Description                                                                    |
|------------------|---------------|------|------------------------|------------------------------------------------|--------------------------------------------------------------------------------|
| payment_no       | VARCHAR(20)   | No   |                        | UQ                                             | Payment reference PY/yy/nnnnnn issued from seq_payment_no.                     |
| agency_id        | UUID          | No   |                        | IDX (1/2)<br>FK agency(id); on delete restrict | Agency or bank that made the payment.                                          |
| submitted_by_id  | UUID          | No   |                        |                                                | User who submitted the payment.                                                |
| method           | PaymentMethod | No   |                        |                                                | BANK_TRANSFER, CHEQUE, CASH_DEPOSIT or ONLINE.                                 |
| bank_name        | VARCHAR(100)  | Yes  |                        |                                                | Paying bank, from master data.                                                 |
| reference_no     | VARCHAR(50)   | No   |                        |                                                | Bank transaction or cheque reference; carried to the FIN interface file.       |
| payment_date     | DATE          | No   |                        |                                                | Date the payment was made.                                                     |
| total_amount     | DECIMAL(14,2) | No   |                        |                                                | Total paid in B\$; must be greater than zero and equal the sum of allocations. |
| status           | PaymentStatus | No   | 'PENDING_VERIFICATION' | IDX (2/2), IDX (1/2)                           | PENDING_VERIFICATION, VERIFIED or REJECTED.                                    |
| remarks          | VARCHAR(500)  | Yes  |                        |                                                | Remarks from the submitter.                                                    |
| verified_by_id   | UUID          | Yes  |                        |                                                | Finance officer who verified or rejected the payment.                          |
| verified_at      | TIMESTAMP(3)  | Yes  |                        |                                                | When the payment was verified or rejected.                                     |
| rejection_reason | VARCHAR(500)  | Yes  |                        |                                                | Reason the payment was rejected.                                               |
| created_at       | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP      | IDX (2/2)                                      | Date and time the row was created (UTC).                                       |

## 2.18 payment\_allocation

How a payment is split across the policies it pays for (bulk payment, AP-38). Model PaymentAllocation; area Billing; primary key id.

| Column     | Data type     | Null | Default                              | Keys / foreign key                                 | Description                                               |
|------------|---------------|------|--------------------------------------|----------------------------------------------------|-----------------------------------------------------------|
| id         | UUID          | No   | UUID v4 generated by the application | PK                                                 | Surrogate primary key.                                    |
| payment_id | UUID          | No   |                                      | UQ (1/2)<br>FK payment(id); on delete cascade      | Payment being allocated.                                  |
| policy_id  | UUID          | No   |                                      | IDX, UQ (2/2)<br>FK policy(id); on delete restrict | Policy the amount pays for.                               |
| amount     | DECIMAL(14,2) | No   |                                      |                                                    | Amount allocated to the policy in B\$; greater than zero. |

## 2.19 receipt

Official e-Receipts issued per policy when a payment is verified (AP-41). Model Receipt; area Billing; primary key id.

| Column      | Data type     | Null | Default                              | Keys / foreign key                       | Description                                                                                   |
|-------------|---------------|------|--------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------|
| id          | UUID          | No   | UUID v4 generated by the application | PK                                       | Surrogate primary key.                                                                        |
| receipt_no  | VARCHAR(20)   | No   |                                      | UQ                                       | Receipt number RC/yy/nnnnnn issued from seq_receipt_no; reconciled with the financial system. |
| payment_id  | UUID          | No   |                                      | FK payment(id); on delete restrict       | Verified payment the receipt was issued from.                                                 |
| policy_id   | UUID          | No   |                                      | IDX<br>FK policy(id); on delete restrict | Policy the receipt is for.                                                                    |
| amount      | DECIMAL(14,2) | No   |                                      |                                          | Amount received for the policy in B\$.                                                        |
| issued_at   | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP                    | IDX                                      | When the receipt was issued; drives the EOD and reconciliation selection.                     |
| document_id | UUID          | Yes  |                                      |                                          | PDF e-Receipt stored as a document.                                                           |

## 2.20 commission

Commission or referral fee accrued per agent and policy at issuance, and marked paid when Finance confirms payment (INT-05). Model Commission; area Billing; primary key id.

| Column       | Data type        | Null | Default                              | Keys / foreign key                                      | Description                                                     |
|--------------|------------------|------|--------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------|
| id           | UUID             | No   | UUID v4 generated by the application | PK                                                      | Surrogate primary key.                                          |
| agent_id     | UUID             | No   |                                      | IDX (1/2), UQ (2/2)<br>FK agent(id); on delete restrict | Agent earning the commission or referral fee.                   |
| policy_id    | UUID             | No   |                                      | UQ (1/2)<br>FK policy(id); on delete restrict           | Policy the commission relates to.                               |
| period       | VARCHAR(7)       | No   |                                      | IDX (2/2)                                               | Accrual period as YYYY-MM.                                      |
| contribution | DECIMAL(14,2)    | No   |                                      |                                                         | Contribution the commission was calculated on, in B\$.          |
| rate         | DECIMAL(6,4)     | No   |                                      |                                                         | Commission rate applied, as a fraction (0.1000 = 10%).          |
| amount       | DECIMAL(14,2)    | No   |                                      |                                                         | Commission amount in B\$.                                       |
| status       | CommissionStatus | No   | 'ACCRUED'                            |                                                         | ACCRUED until Finance confirms payment, then PAID.              |
| paid_at      | TIMESTAMP(3)     | Yes  |                                      |                                                         | Payment date confirmed by Finance through the inbound API.      |
| source       | VARCHAR(20)      | No   |                                      |                                                         | Where the accrual was created; PORTAL for accruals at issuance. |

| Column     | Data type    | Null | Default           | Keys / foreign key | Description                              |
|------------|--------------|------|-------------------|--------------------|------------------------------------------|
| created_at | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP |                    | Date and time the row was created (UTC). |

## 2.21 claim

Claim notifications submitted through the portal (AP-43). Assessment happens in the core system. Model Claim; area Claims; primary key id.

| Column          | Data type     | Null | Default                                | Keys / foreign key                       | Description                                                                             |
|-----------------|---------------|------|----------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------|
| id              | UUID          | No   | UUID v4 generated by the application   | PK                                       | Surrogate primary key.                                                                  |
| claim_no        | VARCHAR(20)   | No   |                                        | UQ                                       | Claim notification reference CL/yy/nnnnnn issued from seq_claim_no.                     |
| policy_id       | UUID          | No   |                                        | IDX<br>FK policy(id); on delete restrict | Policy the claim is notified under.                                                     |
| agent_id        | UUID          | No   |                                        |                                          | Agent who notified the claim (copied from the policy for data scoping).                 |
| agency_id       | UUID          | No   |                                        | IDX (1/2)                                | Agency of the agent (copied from the policy for data scoping).                          |
| claim_type      | VARCHAR(50)   | No   |                                        |                                          | Claim type code from master data: DEATH, TPD, ACCIDENT, MEDICAL, REPATRIATION or OTHER. |
| event_date      | DATE          | No   |                                        |                                          | Date of the insured event.                                                              |
| description     | VARCHAR(2000) | No   |                                        |                                          | Description of the event.                                                               |
| claimed_amount  | DECIMAL(14,2) | Yes  |                                        |                                          | Amount claimed in B\$, where known.                                                     |
| status          | ClaimStatus   | No   | 'SUBMITTED'                            | IDX (2/2)                                | SUBMITTED, UNDER_REVIEW, ACKNOWLEDGED, REJECTED or CLOSED.                              |
| remarks         | VARCHAR(1000) | Yes  |                                        |                                          | Back-office remarks visible to the agent.                                               |
| submitted_by_id | UUID          | No   |                                        |                                          | User who submitted the notification.                                                    |
| created_at      | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP                      |                                          | Date and time the row was created (UTC).                                                |
| updated_at      | TIMESTAMP(3)  | No   | Set by the application on every update |                                          | Date and time the row was last changed (UTC), maintained by the application.            |

## 2.22 document

Metadata of every uploaded or system-generated document. File content is held encrypted in the document store and addressed by storage\_key. Model Document; area Documents; primary key id.

| Column           | Data type         | Null | Default                              | Keys / foreign key | Description                                                                                                                      |
|------------------|-------------------|------|--------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| id               | UUID              | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                                                           |
| owner_type       | DocumentOwnerType | No   |                                      | IDX (1/2)          | Type of record the document belongs to: AGENT, AGENCY, PARTICIPANT, POLICY, PAYMENT, CLAIM, ISSUE or REPORT.                     |
| owner_id         | UUID              | No   |                                      | IDX (2/2)          | Id of the owning record (polymorphic, no foreign key).                                                                           |
| doc_type         | VARCHAR(50)       | No   |                                      |                    | Document type code from master data, e.g. IC_COPY, PROPOSAL_FORM, PAYMENT_PROOF, POLICY_SCHEDULE, RECEIPT, EOD_REPORT, FIN_FILE. |
| file_name        | VARCHAR(255)      | No   |                                      |                    | Sanitised file name; the extension always matches the detected content type.                                                     |
| mime_type        | VARCHAR(100)      | No   |                                      |                    | Content type detected from the file's magic bytes (PDF, PNG, JPEG) or set for generated files.                                   |
| size_bytes       | INTEGER           | No   |                                      |                    | File size in bytes before encryption.                                                                                            |
| storage_key      | VARCHAR(255)      | No   |                                      | UQ                 | Location in the encrypted store, yyyy/mm/<uuid>.bin. Generated by the application; never derived from user input.                |
| sha256           | VARCHAR(64)       | No   |                                      |                    | SHA-256 checksum of the original content, for integrity and duplicate checks.                                                    |
| status           | DocumentStatus    | No   | 'UPLOADED'                           | IDX                | UPLOADED, VERIFIED or REJECTED by a back-office checker.                                                                         |
| expiry_date      | DATE              | Yes  |                                      |                    | Expiry date of the document (e.g. passport, licence) for expiry warnings.                                                        |
| remarks          | VARCHAR(500)      | Yes  |                                      |                    | Verifier remarks, required on rejection.                                                                                         |
| system_generated | BOOLEAN           | No   | false                                |                    | True for documents produced by the system (schedule, receipt, reports).                                                          |
| uploaded_by_id   | UUID              | Yes  |                                      |                    | User who uploaded the file; null for system-generated files and participant e-signatures.                                        |
| verified_by_id   | UUID              | Yes  |                                      |                    | Back-office user who verified or rejected the document.                                                                          |
| verified_at      | TIMESTAMP(3)      | Yes  |                                      |                    | When the document was verified or rejected.                                                                                      |
| created_at       | TIMESTAMP(3)      | No   | CURRENT_TIMESTAMP                    |                    | Date and time the row was created (UTC).                                                                                         |

## 2.23 workflow\_definition

Approval workflow per transaction type (COM-04, BO-16). Inactive definitions approve requests automatically. Model WorkflowDefinition; area Workflow; primary key id.

| Column      | Data type    | Null | Default                                | Keys / foreign key | Description                                                                  |
|-------------|--------------|------|----------------------------------------|--------------------|------------------------------------------------------------------------------|
| id          | UUID         | No   | UUID v4 generated by the application   | PK                 | Surrogate primary key.                                                       |
| type        | ApprovalType | No   |                                        | UQ                 | Transaction type governed by the workflow; one definition per type.          |
| name        | VARCHAR(100) | No   |                                        |                    | Workflow name shown to administrators.                                       |
| description | VARCHAR(500) | Yes  |                                        |                    | What the workflow covers.                                                    |
| active      | BOOLEAN      | No   | true                                   |                    | When false, requests of this type are approved automatically on submission.  |
| updated_at  | TIMESTAMP(3) | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application. |

## 2.24 workflow\_step

Levels of a workflow: permission required to approve the level and optional amount threshold above which the level applies (authority limits). Model WorkflowStep; area Workflow; primary key id.

| Column        | Data type     | Null | Default                              | Keys / foreign key                                        | Description                                                                                                   |
|---------------|---------------|------|--------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| id            | UUID          | No   | UUID v4 generated by the application | PK                                                        | Surrogate primary key.                                                                                        |
| definition_id | UUID          | No   |                                      | UQ (1/2)<br>FK workflow_definition(id); on delete cascade | Workflow the step belongs to.                                                                                 |
| level         | INTEGER       | No   |                                      | UQ (2/2)                                                  | Approval level, starting at 1.                                                                                |
| name          | VARCHAR(100)  | No   |                                      |                                                           | Step name shown in the inbox and approval history.                                                            |
| permission    | VARCHAR(80)   | No   |                                      |                                                           | Permission a checker needs to approve this level.                                                             |
| min_amount    | DECIMAL(14,2) | Yes  |                                      |                                                           | Amount threshold in B\$; the level applies only when the request amount is at or above it. Null means always. |

## 2.25 approval\_request

Maker-checker requests raised for a transaction type, with the proposed change as payload and the approval levels resolved at submission. Model ApprovalRequest; area Workflow; primary key id.

| Column            | Data type      | Null | Default                              | Keys / foreign key | Description                                                                                                                     |
|-------------------|----------------|------|--------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| id                | UUID           | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                                                          |
| request_no        | VARCHAR(20)    | No   |                                      | UQ                 | Request reference RQ/yy/nnnnnn issued from seq_request_no.                                                                      |
| type              | ApprovalType   | No   |                                      | IDX (2/2)          | Transaction type, e.g. AGENT_REGISTRATION, POLICY_REFERRAL, PAYMENT_VERIFICATION.                                               |
| entity_type       | VARCHAR(30)    | No   |                                      | IDX (1/2)          | Type of record the request is about, e.g. Agent, Policy, Payment.                                                               |
| entity_id         | UUID           | No   |                                      | IDX (2/2)          | Id of the record the request is about.                                                                                          |
| summary           | VARCHAR(300)   | No   |                                      |                    | One-line summary shown in the inbox.                                                                                            |
| payload           | JSONB          | No   |                                      |                    | Proposed change or context for the checker, e.g. before and after values of a profile update.                                   |
| amount            | DECIMAL(14,2)  | Yes  |                                      |                    | Amount used to select levels with thresholds, e.g. sum covered.                                                                 |
| status            | ApprovalStatus | No   | 'PENDING'                            | IDX (1/2)          | PENDING, APPROVED, REJECTED or WITHDRAWN.                                                                                       |
| current_level     | INTEGER        | No   | 1                                    |                    | Level awaiting a decision.                                                                                                      |
| total_levels      | INTEGER        | No   |                                      |                    | Number of levels that apply to this request, fixed at submission.                                                               |
| level_permissions | TEXT[]         | Yes  |                                      |                    | Permission required at each level, in order, fixed at submission so a later workflow change does not alter a request in flight. |
| maker_id          | UUID           | No   |                                      | IDX                | User who raised the request. Cannot approve it.                                                                                 |
| maker_name        | VARCHAR(150)   | No   |                                      |                    | Maker's name at the time.                                                                                                       |
| agency_id         | UUID           | Yes  |                                      |                    | Agency the request relates to, for portal visibility.                                                                           |
| submitted_at      | TIMESTAMP(3)   | No   | CURRENT_TIMESTAMP                    |                    | When the request was raised.                                                                                                    |
| decided_at        | TIMESTAMP(3)   | Yes  |                                      |                    | When the request reached its final status.                                                                                      |
| final_remarks     | VARCHAR(1000)  | Yes  |                                      |                    | Remarks of the final decision.                                                                                                  |

## 2.26 approval\_action

Every action on an approval request: submit, approve, reject or withdraw, with actor and remarks (AP-61). Model ApprovalAction; area Workflow; primary key id.

| Column     | Data type          | Null | Default                              | Keys / foreign key                                | Description                              |
|------------|--------------------|------|--------------------------------------|---------------------------------------------------|------------------------------------------|
| id         | UUID               | No   | UUID v4 generated by the application | PK                                                | Surrogate primary key.                   |
| request_id | UUID               | No   |                                      | IDX<br>FK approval_request(id); on delete cascade | Request the action belongs to.           |
| level      | INTEGER            | No   |                                      |                                                   | Level acted on; 0 for submission.        |
| action     | ApprovalActionType | No   |                                      |                                                   | SUBMIT, APPROVE, REJECT or WITHDRAW.     |
| actor_id   | UUID               | No   |                                      |                                                   | User who acted.                          |
| actor_name | VARCHAR(150)       | No   |                                      |                                                   | Actor's name at the time.                |
| remarks    | VARCHAR(1000)      | Yes  |                                      |                                                   | Remarks; mandatory for rejection.        |
| created_at | TIMESTAMP(3)       | No   | CURRENT_TIMESTAMP                    |                                                   | Date and time the row was created (UTC). |

## 2.27 issue

Issues reported from the portal or back-office with priority, assignment and SLA timestamps (AP-55, BO-29 to BO-31). Model Issue; area Issue management; primary key id.

| Column           | Data type     | Null | Default                              | Keys / foreign key | Description                                            |
|------------------|---------------|------|--------------------------------------|--------------------|--------------------------------------------------------|
| id               | UUID          | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                 |
| issue_no         | VARCHAR(20)   | No   |                                      | UQ                 | Issue reference IS/yy/nnnnnn issued from seq_issue_no. |
| title            | VARCHAR(200)  | No   |                                      |                    | Short summary of the issue.                            |
| description      | VARCHAR(4000) | No   |                                      |                    | Full description.                                      |
| category         | VARCHAR(50)   | No   |                                      |                    | Issue category code from master data.                  |
| priority         | IssuePriority | No   |                                      | IDX (2/2)          | CRITICAL, HIGH, MEDIUM or LOW; sets the SLA targets.   |
| status           | IssueStatus   | No   | 'OPEN'                               | IDX (1/2)          | OPEN, ASSIGNED, IN_PROGRESS, RESOLVED or CLOSED.       |
| reported_by_id   | UUID          | No   |                                      | IDX                | User who reported the issue.                           |
| reported_by_name | VARCHAR(150)  | No   |                                      |                    | Reporter's name at the time.                           |
| agency_id        | UUID          | Yes  |                                      |                    | Reporter's agency, for portal users.                   |

| Column             | Data type     | Null | Default                                | Keys / foreign key | Description                                                                  |
|--------------------|---------------|------|----------------------------------------|--------------------|------------------------------------------------------------------------------|
| assigned_to_id     | UUID          | Yes  |                                        | IDX                | Support user the issue is assigned to.                                       |
| assigned_to_name   | VARCHAR(150)  | Yes  |                                        |                    | Assignee's name at the time.                                                 |
| assigned_team      | VARCHAR(50)   | Yes  |                                        |                    | Support team the issue is assigned to.                                       |
| response_due_at    | TIMESTAMP(3)  | No   |                                        |                    | Response target: creation time plus the response hours for the priority.     |
| resolution_due_at  | TIMESTAMP(3)  | No   |                                        |                    | Resolution target: creation time plus the resolution hours for the priority. |
| first_responded_at | TIMESTAMP(3)  | Yes  |                                        |                    | When support first responded.                                                |
| resolved_at        | TIMESTAMP(3)  | Yes  |                                        |                    | When the issue was resolved.                                                 |
| closed_at          | TIMESTAMP(3)  | Yes  |                                        |                    | When the issue was closed.                                                   |
| sla_breached       | BOOLEAN       | No   | false                                  |                    | Set by the SLA monitor when a target is missed; support is alerted once.     |
| resolution         | VARCHAR(2000) | Yes  |                                        |                    | Resolution notes.                                                            |
| created_at         | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP                      |                    | Date and time the row was created (UTC).                                     |
| updated_at         | TIMESTAMP(3)  | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application. |

## 2.28 issue\_comment

Comments on an issue. Internal comments are visible only to back-office staff. Model IssueComment; area Issue management; primary key id.

| Column      | Data type     | Null | Default                              | Keys / foreign key                     | Description                              |
|-------------|---------------|------|--------------------------------------|----------------------------------------|------------------------------------------|
| id          | UUID          | No   | UUID v4 generated by the application | PK                                     | Surrogate primary key.                   |
| issue_id    | UUID          | No   |                                      | IDX<br>FK issue(id); on delete cascade | Issue the comment belongs to.            |
| author_id   | UUID          | No   |                                      |                                        | User who wrote the comment.              |
| author_name | VARCHAR(150)  | No   |                                      |                                        | Author's name at the time.               |
| body        | VARCHAR(4000) | No   |                                      |                                        | Comment text.                            |
| internal    | BOOLEAN       | No   | false                                |                                        | True for back-office-only comments.      |
| created_at  | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP                    |                                        | Date and time the row was created (UTC). |

## 2.29 notification

In-app, e-mail and SMS notifications. E-mail and SMS rows are delivered asynchronously by the notification dispatcher with retry. Model Notification; area Notifications; primary key id.

| Column         | Data type           | Null | Default                              | Keys / foreign key                              | Description                                                                                                                                                    |
|----------------|---------------------|------|--------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id             | UUID                | No   | UUID v4 generated by the application | PK                                              | Surrogate primary key.                                                                                                                                         |
| user_id        | UUID                | Yes  |                                      | IDX (1/3)<br>FK app_user(id); on delete cascade | Recipient user; null for external recipients such as participants and report subscribers.                                                                      |
| channel        | NotificationChannel | No   |                                      | IDX (2/3), IDX (2/2)                            | IN_APP, EMAIL or SMS.                                                                                                                                          |
| recipient      | VARCHAR(254)        | No   |                                      |                                                 | E-mail address, mobile number or user id, depending on the channel.                                                                                            |
| event_type     | VARCHAR(50)         | No   |                                      |                                                 | Event that caused the notification, e.g. POLICY_ISSUED, APPROVAL_REQUIRED, AGENCY_BLOCKED.                                                                     |
| subject        | VARCHAR(200)        | No   |                                      |                                                 | Subject line or in-app title.                                                                                                                                  |
| body           | VARCHAR(4000)       | No   |                                      |                                                 | Message text. For sensitive messages (e.g. e-signature links) replaced with a placeholder after delivery. <b>Removed after delivery when sensitive = true.</b> |
| link           | VARCHAR(300)        | Yes  |                                      |                                                 | Application path the in-app notification opens.                                                                                                                |
| attachment_ids | TEXT[]              | Yes  | Empty array                          |                                                 | Documents attached to the e-mail.                                                                                                                              |
| sensitive      | BOOLEAN             | No   | false                                |                                                 | True when the body must be removed once delivered.                                                                                                             |
| status         | DeliveryStatus      | No   | 'PENDING'                            | IDX (1/2)                                       | PENDING, SENT or FAILED (after notification.max_attempts attempts).                                                                                            |
| attempts       | INTEGER             | No   | 0                                    |                                                 | Delivery attempts so far.                                                                                                                                      |
| last_error     | VARCHAR(1000)       | Yes  |                                      |                                                 | Error from the last failed attempt.                                                                                                                            |
| read_at        | TIMESTAMP(3)        | Yes  |                                      | IDX (3/3)                                       | When the user read an in-app notification.                                                                                                                     |
| sent_at        | TIMESTAMP(3)        | Yes  |                                      |                                                 | When the e-mail or SMS was accepted by the gateway.                                                                                                            |
| created_at     | TIMESTAMP(3)        | No   | CURRENT_TIMESTAMP                    |                                                 | Date and time the row was created (UTC).                                                                                                                       |

## 2.30 audit\_log

Append-only audit trail of security and business events with before and after values (BO-26 to BO-28, COM-03). A trigger rejects UPDATE and DELETE. Model AuditLog; area Audit trail; primary key id.

| Column | Data type | Null | Default                           | Keys / foreign key | Description                                                  |
|--------|-----------|------|-----------------------------------|--------------------|--------------------------------------------------------------|
| id     | BIGSERIAL | No   | Next value of the column sequence | PK                 | Sequential primary key; gives a total order of audit events. |

| Column         | Data type    | Null | Default           | Keys / foreign key | Description                                                                                                            |
|----------------|--------------|------|-------------------|--------------------|------------------------------------------------------------------------------------------------------------------------|
| occurred_at    | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP | IDX, IDX (2/2)     | When the event happened (UTC).                                                                                         |
| actor_id       | UUID         | Yes  |                   | IDX (1/2)          | User who performed the action; null for system jobs and inbound integrations.                                          |
| actor_name     | VARCHAR(150) | Yes  |                   |                    | Username of the actor, or 'system'.                                                                                    |
| action         | VARCHAR(60)  | No   |                   | IDX                | Event code, e.g. LOGIN, LOGIN_FAILED, ACCOUNT_LOCKED, APPROVAL_DECIDED, PRODUCT_UPDATED, AGENCY_BLOCKED.               |
| entity_type    | VARCHAR(40)  | No   |                   | IDX (1/2)          | Type of record affected.                                                                                               |
| entity_id      | VARCHAR(64)  | Yes  |                   | IDX (2/2)          | Id of the record affected.                                                                                             |
| before         | JSONB        | Yes  |                   |                    | Values before the change. Passwords, hashes, encrypted identifiers, tokens and storage keys are written as [REDACTED]. |
| after          | JSONB        | Yes  |                   |                    | Values after the change, redacted in the same way.                                                                     |
| ip_address     | VARCHAR(64)  | Yes  |                   |                    | Client IP address as seen through the reverse proxy.                                                                   |
| user_agent     | VARCHAR(300) | Yes  |                   |                    | Client browser user agent, truncated to 300 characters.                                                                |
| correlation_id | VARCHAR(64)  | Yes  |                   |                    | Request correlation id (X-Request-Id); links the event to application logs.                                            |

## 2.31 config\_parameter

Business parameters administrators can change without a release: password policy, session timeouts, grace period, SLA hours and similar (BO-33, COM-09). Model ConfigParameter; area Configuration and master data; primary key key.

| Column        | Data type     | Null | Default                                | Keys / foreign key | Description                                                                  |
|---------------|---------------|------|----------------------------------------|--------------------|------------------------------------------------------------------------------|
| key           | VARCHAR(80)   | No   |                                        | PK                 | Parameter key, e.g. security.lockout.max_attempts.                           |
| value         | VARCHAR(1000) | No   |                                        |                    | Current value as text, interpreted according to value_type.                  |
| value_type    | ParameterType | No   |                                        |                    | STRING, INTEGER, DECIMAL or BOOLEAN.                                         |
| category      | VARCHAR(50)   | No   |                                        |                    | Grouping shown in the back-office, e.g. Security, Billing, Issue SLA.        |
| description   | VARCHAR(500)  | No   |                                        |                    | What the parameter controls.                                                 |
| min_value     | DECIMAL(14,2) | Yes  |                                        |                    | Lowest value accepted for numeric parameters.                                |
| max_value     | DECIMAL(14,2) | Yes  |                                        |                    | Highest value accepted for numeric parameters.                               |
| updated_by_id | UUID          | Yes  |                                        |                    | Administrator who last changed the value.                                    |
| updated_at    | TIMESTAMP(3)  | No   | Set by the application on every update |                    | Date and time the row was last changed (UTC), maintained by the application. |

## 2.32 code\_item

Master data code lists (document types, relationships, nationalities, districts, banks, occupations, claim types, issue categories and others) maintained in the back-office (BO-32). Model CodeItem; area Configuration and master data; primary key id.

| Column     | Data type    | Null | Default                              | Keys / foreign key | Description                                                                      |
|------------|--------------|------|--------------------------------------|--------------------|----------------------------------------------------------------------------------|
| id         | UUID         | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                           |
| category   | VARCHAR(50)  | No   |                                      | UQ (1/2)           | Code list name, e.g. DOCUMENT_TYPE, RELATIONSHIP, DISTRICT, BANK, CLAIM_TYPE.    |
| code       | VARCHAR(50)  | No   |                                      | UQ (2/2)           | Stable code stored on business records.                                          |
| label      | VARCHAR(150) | No   |                                      |                    | Label shown to users.                                                            |
| active     | BOOLEAN      | No   | true                                 |                    | Inactive codes are hidden from new entries but remain valid on existing records. |
| sort_order | INTEGER      | No   | 0                                    |                    | Display order within the list.                                                   |

## 2.33 outbox\_message

Transactional outbox for messages to the core and financial systems. Written in the same transaction as the business change and delivered asynchronously with retry. Model OutboxMessage; area Integration; primary key id.

| Column          | Data type         | Null | Default                              | Keys / foreign key | Description                                                                                       |
|-----------------|-------------------|------|--------------------------------------|--------------------|---------------------------------------------------------------------------------------------------|
| id              | UUID              | No   | UUID v4 generated by the application | PK                 | Primary key; also sent as the Idempotency-Key header so the receiver can discard duplicates.      |
| system          | IntegrationSystem | No   |                                      | IDX (1/2)          | Target system: CORE or FINANCE.                                                                   |
| operation       | VARCHAR(60)       | No   |                                      |                    | Operation code from the interface specification, e.g. POLICY_ISSUED, RECEIPT_POSTED, EOD_POSTING. |
| payload         | JSONB             | No   |                                      |                    | JSON message body.                                                                                |
| status          | OutboxStatus      | No   | 'PENDING'                            | IDX (1/2)          | PENDING, SENT, FAILED (retry scheduled) or DEAD (attempts exhausted or not retryable).            |
| attempts        | INTEGER           | No   | 0                                    |                    | Delivery attempts so far.                                                                         |
| next_attempt_at | TIMESTAMP(3)      | No   | CURRENT_TIMESTAMP                    | IDX (2/2)          | Earliest time of the next attempt (exponential back-off, capped at 60 minutes).                   |
| last_error      | VARCHAR(1000)     | Yes  |                                      |                    | Error from the last failed attempt.                                                               |
| aggregate_type  | VARCHAR(30)       | Yes  |                                      |                    | Type of business record that produced the message, e.g. Policy, Receipt.                          |
| aggregate_id    | VARCHAR(64)       | Yes  |                                      |                    | Id of the business record.                                                                        |

| Column         | Data type    | Null | Default           | Keys / foreign key | Description                                              |
|----------------|--------------|------|-------------------|--------------------|----------------------------------------------------------|
| correlation_id | VARCHAR(64)  | Yes  |                   |                    | Correlation id of the request that produced the message. |
| created_at     | TIMESTAMP(3) | No   | CURRENT_TIMESTAMP | IDX (2/2)          | Date and time the row was created (UTC).                 |
| processed_at   | TIMESTAMP(3) | Yes  |                   |                    | When the message was delivered.                          |

## 2.34 integration\_log

One row per integration call, outbound or inbound, with outcome and duration (INT-11, INT-14). Model IntegrationLog; area Integration; primary key id.

| Column           | Data type            | Null | Default                              | Keys / foreign key | Description                                                                                                                      |
|------------------|----------------------|------|--------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| id               | UUID                 | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                                                                           |
| system           | IntegrationSystem    | No   |                                      | IDX (1/2)          | CORE, FINANCE, AML, EMAIL, SMS or DIRECTORY.                                                                                     |
| operation        | VARCHAR(60)          | No   |                                      |                    | Operation or notification event code.                                                                                            |
| direction        | IntegrationDirection | No   |                                      |                    | OUTBOUND (portal calls the system) or INBOUND (system calls the portal).                                                         |
| success          | BOOLEAN              | No   |                                      | IDX (1/2)          | Whether the call succeeded.                                                                                                      |
| duration_ms      | INTEGER              | No   |                                      |                    | Call duration in milliseconds.                                                                                                   |
| reference        | VARCHAR(100)         | Yes  |                                      |                    | Reference returned by the receiver, or the business reference of an inbound call. Simulated deliveries are marked '(simulated)'. |
| request_summary  | VARCHAR(1000)        | Yes  |                                      |                    | Short description of the request, e.g. aggregate type and id. Never the full payload.                                            |
| response_summary | VARCHAR(1000)        | Yes  |                                      |                    | Short description of the response.                                                                                               |
| error_message    | VARCHAR(1000)        | Yes  |                                      |                    | Error text for failed calls.                                                                                                     |
| outbox_id        | UUID                 | Yes  |                                      |                    | Outbox message delivered by this call, if any.                                                                                   |
| created_at       | TIMESTAMP(3)         | No   | CURRENT_TIMESTAMP                    | IDX (2/2)          | Date and time the row was created (UTC).                                                                                         |

## 2.35 reconciliation\_run

Results of daily reconciliation between receipts issued by the portal and receipts posted by the financial system (INT-15). Model ReconciliationRun; area Integration; primary key id.

| Column | Data type | Null | Default                              | Keys / foreign key | Description            |
|--------|-----------|------|--------------------------------------|--------------------|------------------------|
| id     | UUID      | No   | UUID v4 generated by the application | PK                 | Surrogate primary key. |

| Column          | Data type            | Null | Default           | Keys / foreign key | Description                                               |
|-----------------|----------------------|------|-------------------|--------------------|-----------------------------------------------------------|
| business_date   | DATE                 | No   |                   | IDX (1/2)          | Business date reconciled.                                 |
| system          | IntegrationSystem    | No   |                   | IDX (2/2)          | System reconciled against; FINANCE.                       |
| expected_count  | INTEGER              | No   |                   |                    | Receipts issued by the portal on the date.                |
| matched_count   | INTEGER              | No   |                   |                    | Receipts also reported as posted by the financial system. |
| expected_amount | DECIMAL(14,2)        | No   |                   |                    | Total of receipts issued, in B\$.                         |
| matched_amount  | DECIMAL(14,2)        | No   |                   |                    | Total of matched receipts, in B\$.                        |
| status          | ReconciliationStatus | No   |                   |                    | MATCHED or MISMATCH.                                      |
| details         | JSONB                | No   |                   |                    | Unmatched receipt numbers for investigation.              |
| created_at      | TIMESTAMP(3)         | No   | CURRENT_TIMESTAMP |                    | Date and time the row was created (UTC).                  |

## 2.36 eod\_run

End-of-day runs per business date with totals and links to the EOD report and FIN interface file. Model EodRun; area Integration; primary key id.

| Column               | Data type     | Null | Default                              | Keys / foreign key | Description                                                                |
|----------------------|---------------|------|--------------------------------------|--------------------|----------------------------------------------------------------------------|
| id                   | UUID          | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                     |
| business_date        | DATE          | No   |                                      | UQ                 | Business date processed; one run per date (a re-run updates the same row). |
| status               | JobStatus     | No   |                                      |                    | RUNNING, COMPLETED or FAILED.                                              |
| started_at           | TIMESTAMP(3)  | No   | CURRENT_TIMESTAMP                    |                    | When the run started.                                                      |
| finished_at          | TIMESTAMP(3)  | Yes  |                                      |                    | When the run finished.                                                     |
| policies_issued      | INTEGER       | No   | 0                                    |                    | Policies issued on the date.                                               |
| receipts_issued      | INTEGER       | No   | 0                                    |                    | Receipts issued on the date.                                               |
| total_contribution   | DECIMAL(14,2) | No   | 0                                    |                    | Total contribution of policies issued, in B\$.                             |
| total_receipts       | DECIMAL(14,2) | No   | 0                                    |                    | Total of receipts issued, in B\$.                                          |
| report_document_id   | UUID          | Yes  |                                      |                    | EOD issuance report (Excel) stored as a document.                          |
| fin_file_document_id | UUID          | Yes  |                                      |                    | FIN interface file (CSV) stored as a document.                             |
| error_message        | VARCHAR(1000) | Yes  |                                      |                    | Error text of a failed run.                                                |

| Column       | Data type    | Null | Default | Keys / foreign key | Description                                                          |
|--------------|--------------|------|---------|--------------------|----------------------------------------------------------------------|
| triggered_by | VARCHAR(150) | No   |         |                    | 'Scheduler' or the name of the user who ran it from the back-office. |

## 2.37 report\_schedule

Scheduled reports e-mailed to recipients daily, weekly or monthly (BO-25). Model ReportSchedule; area Integration; primary key id.

| Column        | Data type       | Null | Default                              | Keys / foreign key | Description                                                                       |
|---------------|-----------------|------|--------------------------------------|--------------------|-----------------------------------------------------------------------------------|
| id            | UUID            | No   | UUID v4 generated by the application | PK                 | Surrogate primary key.                                                            |
| report_code   | VARCHAR(50)     | No   |                                      |                    | Report from the report catalogue, e.g. POLICY_REGISTER, COLLECTIONS.              |
| name          | VARCHAR(150)    | No   |                                      |                    | Schedule name used in the e-mail subject.                                         |
| frequency     | ReportFrequency | No   |                                      |                    | DAILY (07:00), WEEKLY (Monday) or MONTHLY (1st), Brunei time.                     |
| format        | ExportFormat    | No   |                                      |                    | XLSX, CSV or PDF.                                                                 |
| recipients    | TEXT[]          | Yes  |                                      |                    | E-mail addresses that receive the report.                                         |
| filters       | JSONB           | No   |                                      |                    | Report period, e.g. PREVIOUS_DAY, PREVIOUS_7_DAYS, PREVIOUS_MONTH, MONTH_TO_DATE. |
| active        | BOOLEAN         | No   | true                                 | IDX (1/2)          | Paused schedules are not run.                                                     |
| last_run_at   | TIMESTAMP(3)    | Yes  |                                      |                    | When the schedule last ran.                                                       |
| next_run_at   | TIMESTAMP(3)    | No   |                                      | IDX (2/2)          | When the schedule runs next.                                                      |
| created_by_id | UUID            | No   |                                      |                    | User who created the schedule.                                                    |
| created_at    | TIMESTAMP(3)    | No   | CURRENT_TIMESTAMP                    |                    | Date and time the row was created (UTC).                                          |

### 3. Relationships

Every foreign key in the database. The parent row must exist before a child row can reference it. RESTRICT prevents deleting a parent that still has children; CASCADE deletes dependent rows with the parent; SET NULL clears the reference.

| Parent      | Child              | Child column    | Cardinality | On delete | Constraint                         |
|-------------|--------------------|-----------------|-------------|-----------|------------------------------------|
| agent       | app_user           | agent_id        | 1 : 0..1    | SET NULL  | app_user_agent_id_fkey             |
| app_user    | password_history   | user_id         | 1 : 0..n    | CASCADE   | password_history_user_id_fkey      |
| role        | role_permission    | role_id         | 1 : 0..n    | CASCADE   | role_permission_role_id_fkey       |
| app_user    | user_role          | user_id         | 1 : 0..n    | CASCADE   | user_role_user_id_fkey             |
| role        | user_role          | role_id         | 1 : 0..n    | CASCADE   | user_role_role_id_fkey             |
| agency      | agent              | agency_id       | 1 : 0..n    | RESTRICT  | agent_agency_id_fkey               |
| agent       | agent              | parent_agent_id | 1 : 0..n    | SET NULL  | agent_parent_agent_id_fkey         |
| product     | policy             | product_id      | 1 : 0..n    | RESTRICT  | policy_product_id_fkey             |
| participant | policy             | participant_id  | 1 : 0..n    | RESTRICT  | policy_participant_id_fkey         |
| agent       | policy             | agent_id        | 1 : 0..n    | RESTRICT  | policy_agent_id_fkey               |
| agency      | policy             | agency_id       | 1 : 0..n    | RESTRICT  | policy_agency_id_fkey              |
| policy      | policy             | renewal_of_id   | 1 : 0..n    | SET NULL  | policy_renewal_of_id_fkey          |
| policy      | nominee            | policy_id       | 1 : 0..n    | CASCADE   | nominee_policy_id_fkey             |
| policy      | policy_event       | policy_id       | 1 : 0..n    | CASCADE   | policy_event_policy_id_fkey        |
| policy      | signature_request  | policy_id       | 1 : 0..n    | CASCADE   | signature_request_policy_id_fkey   |
| agency      | payment            | agency_id       | 1 : 0..n    | RESTRICT  | payment_agency_id_fkey             |
| payment     | payment_allocation | payment_id      | 1 : 0..n    | CASCADE   | payment_allocation_payment_id_fkey |
| policy      | payment_allocation | policy_id       | 1 : 0..n    | RESTRICT  | payment_allocation_policy_id_fkey  |
| payment     | receipt            | payment_id      | 1 : 0..n    | RESTRICT  | receipt_payment_id_fkey            |
| policy      | receipt            | policy_id       | 1 : 0..n    | RESTRICT  | receipt_policy_id_fkey             |
| agent       | commission         | agent_id        | 1 : 0..n    | RESTRICT  | commission_agent_id_fkey           |
| policy      | commission         | policy_id       | 1 : 0..n    | RESTRICT  | commission_policy_id_fkey          |
| policy      | claim              | policy_id       | 1 : 0..n    | RESTRICT  | claim_policy_id_fkey               |

| Parent              | Child           | Child column  | Cardinality | On delete | Constraint                       |
|---------------------|-----------------|---------------|-------------|-----------|----------------------------------|
| workflow_definition | workflow_step   | definition_id | 1 : 0..n    | CASCADE   | workflow_step_definition_id_fkey |
| approval_request    | approval_action | request_id    | 1 : 0..n    | CASCADE   | approval_action_request_id_fkey  |
| issue               | issue_comment   | issue_id      | 1 : 0..n    | CASCADE   | issue_comment_issue_id_fkey      |
| app_user            | notification    | user_id       | 1 : 0..n    | CASCADE   | notification_user_id_fkey        |

Table 3: Foreign keys

Polymorphic references are deliberately not foreign keys: document.owner\_type/owner\_id, approval\_request.entity\_type/entity\_id, aml\_screening.subject\_type/subject\_id, outbox\_message.aggregate\_type/aggregate\_id and audit\_log.entity\_type/entity\_id. The application checks these references, and history tables keep the actor name so records remain readable after a user is deactivated.

## 4. Indexes

Indexes created by the migrations, grouped by table. Composite indexes serve the most frequent list screens: for example policy (agency\_id, status) for the portal policy list and outbox\_message (status, next\_attempt\_at) for the dispatcher.

| Table               | Index                                     | Columns                  | Unique | Purpose                                                    |
|---------------------|-------------------------------------------|--------------------------|--------|------------------------------------------------------------|
| app_user            | app_user_pkey                             | id                       | Yes    | Primary key                                                |
| app_user            | app_user_agent_id_key                     | agent_id                 | Yes    | Enforces uniqueness of agent_id                            |
| app_user            | app_user_user_type_status_idx             | user_type, status        | No     | Supports filtering and sorting on user_type, status        |
| app_user            | app_user_username_key                     | username                 | Yes    | Enforces uniqueness of username                            |
| password_history    | password_history_pkey                     | id                       | Yes    | Primary key                                                |
| password_history    | password_history_user_id_created_at_idx   | user_id, created_at      | No     | Finds child rows of a parent; supports the foreign key     |
| role                | role_pkey                                 | id                       | Yes    | Primary key                                                |
| role                | role_code_key                             | code                     | Yes    | Enforces uniqueness of code                                |
| role_permission     | role_permission_pkey                      | role_id, permission      | Yes    | Primary key                                                |
| user_role           | user_role_pkey                            | user_id, role_id         | Yes    | Primary key                                                |
| user_session        | user_session_pkey                         | sid                      | Yes    | Primary key                                                |
| user_session        | user_session_expire_idx                   | expire                   | No     | Supports filtering and sorting on expire                   |
| agency              | agency_pkey                               | id                       | Yes    | Primary key                                                |
| agency              | agency_code_key                           | code                     | Yes    | Enforces uniqueness of code                                |
| agent               | agent_pkey                                | id                       | Yes    | Primary key                                                |
| agent               | agent_agency_id_status_idx                | agency_id, status        | No     | Finds child rows of a parent; supports the foreign key     |
| agent               | agent_agent_code_key                      | agent_code               | Yes    | Enforces uniqueness of agent_code                          |
| agent               | agent_id_type_id_number_hash_key          | id_type, id_number_hash  | Yes    | Enforces uniqueness of id_type + id_number_hash            |
| agent               | agent_parent_agent_id_idx                 | parent_agent_id          | No     | Finds child rows of a parent; supports the foreign key     |
| participant         | participant_pkey                          | id                       | Yes    | Primary key                                                |
| participant         | participant_created_by_agency_id_idx      | created_by_agency_id     | No     | Supports filtering and sorting on created_by_agency_id     |
| participant         | participant_full_name_idx                 | full_name                | No     | Supports filtering and sorting on full_name                |
| participant         | participant_id_type_id_number_hash_key    | id_type, id_number_hash  | Yes    | Enforces uniqueness of id_type + id_number_hash            |
| participant         | participant_participant_no_key            | participant_no           | Yes    | Enforces uniqueness of participant_no                      |
| aml_screening       | aml_screening_pkey                        | id                       | Yes    | Primary key                                                |
| aml_screening       | aml_screening_status_idx                  | status                   | No     | Supports filtering and sorting on status                   |
| aml_screening       | aml_screening_subject_type_subject_id_idx | subject_type, subject_id | No     | Supports filtering and sorting on subject_type, subject_id |
| aml_watchlist_entry | aml_watchlist_entry_pkey                  | id                       | Yes    | Primary key                                                |
| aml_watchlist_entry | aml_watchlist_entry_normalised_idx        | normalised               | No     | Supports filtering and sorting on normalised               |

| Table              | Index                                             | Columns                                  | Unique | Purpose                                                                    |
|--------------------|---------------------------------------------------|------------------------------------------|--------|----------------------------------------------------------------------------|
| product            | product_pkey                                      | id                                       | Yes    | Primary key                                                                |
| product            | product_code_key                                  | code                                     | Yes    | Enforces uniqueness of code                                                |
| policy             | policy_pkey                                       | id                                       | Yes    | Primary key                                                                |
| policy             | policy_agency_id_status_idx                       | agency_id, status                        | No     | Finds child rows of a parent; supports the foreign key                     |
| policy             | policy_agent_id_status_idx                        | agent_id, status                         | No     | Finds child rows of a parent; supports the foreign key                     |
| policy             | policy_issued_at_idx                              | issued_at                                | No     | Supports filtering and sorting on issued_at                                |
| policy             | policy_participant_id_idx                         | participant_id                           | No     | Finds child rows of a parent; supports the foreign key                     |
| policy             | policy_policy_no_key                              | policy_no                                | Yes    | Enforces uniqueness of policy_no                                           |
| policy             | policy_quotation_no_key                           | quotation_no                             | Yes    | Enforces uniqueness of quotation_no                                        |
| policy             | policy_status_payment_status_payment_due_date_idx | status, payment_status, payment_due_date | No     | Supports filtering and sorting on status, payment_status, payment_due_date |
| nominee            | nominee_pkey                                      | id                                       | Yes    | Primary key                                                                |
| nominee            | nominee_policy_id_idx                             | policy_id                                | No     | Finds child rows of a parent; supports the foreign key                     |
| policy_event       | policy_event_pkey                                 | id                                       | Yes    | Primary key                                                                |
| policy_event       | policy_event_policy_id_created_at_idx             | policy_id, created_at                    | No     | Finds child rows of a parent; supports the foreign key                     |
| signature_request  | signature_request_pkey                            | id                                       | Yes    | Primary key                                                                |
| signature_request  | signature_request_policy_id_idx                   | policy_id                                | No     | Finds child rows of a parent; supports the foreign key                     |
| signature_request  | signature_request_token_hash_key                  | token_hash                               | Yes    | Enforces uniqueness of token_hash                                          |
| payment            | payment_pkey                                      | id                                       | Yes    | Primary key                                                                |
| payment            | payment_agency_id_status_idx                      | agency_id, status                        | No     | Finds child rows of a parent; supports the foreign key                     |
| payment            | payment_payment_no_key                            | payment_no                               | Yes    | Enforces uniqueness of payment_no                                          |
| payment            | payment_status_created_at_idx                     | status, created_at                       | No     | Supports filtering and sorting on status, created_at                       |
| payment_allocation | payment_allocation_pkey                           | id                                       | Yes    | Primary key                                                                |
| payment_allocation | payment_allocation_payment_id_policy_id_key       | payment_id, policy_id                    | Yes    | Enforces uniqueness of payment_id + policy_id                              |
| payment_allocation | payment_allocation_policy_id_idx                  | policy_id                                | No     | Finds child rows of a parent; supports the foreign key                     |
| receipt            | receipt_pkey                                      | id                                       | Yes    | Primary key                                                                |
| receipt            | receipt_issued_at_idx                             | issued_at                                | No     | Supports filtering and sorting on issued_at                                |
| receipt            | receipt_policy_id_idx                             | policy_id                                | No     | Finds child rows of a parent; supports the foreign key                     |
| receipt            | receipt_receipt_no_key                            | receipt_no                               | Yes    | Enforces uniqueness of receipt_no                                          |
| commission         | commission_pkey                                   | id                                       | Yes    | Primary key                                                                |
| commission         | commission_agent_id_period_idx                    | agent_id, period                         | No     | Finds child rows of a parent; supports the foreign key                     |
| commission         | commission_policy_id_agent_id_key                 | policy_id, agent_id                      | Yes    | Enforces uniqueness of policy_id + agent_id                                |
| claim              | claim_pkey                                        | id                                       | Yes    | Primary key                                                                |
| claim              | claim_agency_id_status_idx                        | agency_id, status                        | No     | Supports filtering and sorting on agency_id, status                        |
| claim              | claim_claim_no_key                                | claim_no                                 | Yes    | Enforces uniqueness of claim_no                                            |

| Table               | Index                                      | Columns                   | Unique | Purpose                                                  |
|---------------------|--------------------------------------------|---------------------------|--------|----------------------------------------------------------|
| claim               | claim_policy_id_idx                        | policy_id                 | No     | Finds child rows of a parent; supports the foreign key   |
| document            | document_pkey                              | id                        | Yes    | Primary key                                              |
| document            | document_owner_type_owner_id_idx           | owner_type, owner_id      | No     | Supports filtering and sorting on owner_type, owner_id   |
| document            | document_status_idx                        | status                    | No     | Supports filtering and sorting on status                 |
| document            | document_storage_key_key                   | storage_key               | Yes    | Enforces uniqueness of storage_key                       |
| workflow_definition | workflow_definition_pkey                   | id                        | Yes    | Primary key                                              |
| workflow_definition | workflow_definition_type_key               | type                      | Yes    | Enforces uniqueness of type                              |
| workflow_step       | workflow_step_pkey                         | id                        | Yes    | Primary key                                              |
| workflow_step       | workflow_step_definition_id_level_key      | definition_id, level      | Yes    | Enforces uniqueness of definition_id + level             |
| approval_request    | approval_request_pkey                      | id                        | Yes    | Primary key                                              |
| approval_request    | approval_request_entity_type_entity_id_idx | entity_type, entity_id    | No     | Supports filtering and sorting on entity_type, entity_id |
| approval_request    | approval_request_maker_id_idx              | maker_id                  | No     | Supports filtering and sorting on maker_id               |
| approval_request    | approval_request_request_no_key            | request_no                | Yes    | Enforces uniqueness of request_no                        |
| approval_request    | approval_request_status_type_idx           | status, type              | No     | Supports filtering and sorting on status, type           |
| approval_action     | approval_action_pkey                       | id                        | Yes    | Primary key                                              |
| approval_action     | approval_action_request_id_idx             | request_id                | No     | Finds child rows of a parent; supports the foreign key   |
| issue               | issue_pkey                                 | id                        | Yes    | Primary key                                              |
| issue               | issue_assigned_to_id_idx                   | assigned_to_id            | No     | Supports filtering and sorting on assigned_to_id         |
| issue               | issue_issue_no_key                         | issue_no                  | Yes    | Enforces uniqueness of issue_no                          |
| issue               | issue_reported_by_id_idx                   | reported_by_id            | No     | Supports filtering and sorting on reported_by_id         |
| issue               | issue_status_priority_idx                  | status, priority          | No     | Supports filtering and sorting on status, priority       |
| issue_comment       | issue_comment_pkey                         | id                        | Yes    | Primary key                                              |
| issue_comment       | issue_comment_issue_id_idx                 | issue_id                  | No     | Finds child rows of a parent; supports the foreign key   |
| notification        | notification_pkey                          | id                        | Yes    | Primary key                                              |
| notification        | notification_status_channel_idx            | status, channel           | No     | Supports filtering and sorting on status, channel        |
| notification        | notification_user_id_channel_read_at_idx   | user_id, channel, read_at | No     | Finds child rows of a parent; supports the foreign key   |
| audit_log           | audit_log_pkey                             | id                        | Yes    | Primary key                                              |
| audit_log           | audit_log_action_idx                       | action                    | No     | Supports filtering and sorting on action                 |
| audit_log           | audit_log_actor_id_occurred_at_idx         | actor_id, occurred_at     | No     | Supports filtering and sorting on actor_id, occurred_at  |
| audit_log           | audit_log_entity_type_entity_id_idx        | entity_type, entity_id    | No     | Supports filtering and sorting on entity_type, entity_id |
| audit_log           | audit_log_occurred_at_idx                  | occurred_at               | No     | Supports filtering and sorting on occurred_at            |
| config_parameter    | config_parameter_pkey                      | key                       | Yes    | Primary key                                              |
| code_item           | code_item_pkey                             | id                        | Yes    | Primary key                                              |
| code_item           | code_item_category_code_key                | category, code            | Yes    | Enforces uniqueness of category + code                   |
| outbox_message      | outbox_message_pkey                        | id                        | Yes    | Primary key                                              |

| Table              | Index                                       | Columns                 | Unique | Purpose                                                   |
|--------------------|---------------------------------------------|-------------------------|--------|-----------------------------------------------------------|
| outbox_message     | outbox_message_status_next_attempt_at_idx   | status, next_attempt_at | No     | Supports filtering and sorting on status, next_attempt_at |
| outbox_message     | outbox_message_system_created_at_idx        | system, created_at      | No     | Supports filtering and sorting on system, created_at      |
| integration_log    | integration_log_pkey                        | id                      | Yes    | Primary key                                               |
| integration_log    | integration_log_success_created_at_idx      | success, created_at     | No     | Supports filtering and sorting on success, created_at     |
| integration_log    | integration_log_system_created_at_idx       | system, created_at      | No     | Supports filtering and sorting on system, created_at      |
| reconciliation_run | reconciliation_run_pkey                     | id                      | Yes    | Primary key                                               |
| reconciliation_run | reconciliation_run_business_date_system_idx | business_date, system   | No     | Supports filtering and sorting on business_date, system   |
| eod_run            | eod_run_pkey                                | id                      | Yes    | Primary key                                               |
| eod_run            | eod_run_business_date_key                   | business_date           | Yes    | Enforces uniqueness of business_date                      |
| report_schedule    | report_schedule_pkey                        | id                      | Yes    | Primary key                                               |
| report_schedule    | report_schedule_active_next_run_at_idx      | active, next_run_at     | No     | Supports filtering and sorting on active, next_run_at     |

Table 4: Indexes

## 5. Enumerations

Enumerations are PostgreSQL ENUM types. Adding a value requires a migration; values are never removed while rows use them.

### UserType: kind of user account.

| Value | Meaning                                     |
|-------|---------------------------------------------|
| AGENT | Agency agent or sub-agent using the portal. |
| BANCA | Bank officer using the portal.              |
| STAFF | IIFT staff using the back-office.           |

Used by: app\_user.user\_type

### UserStatus: state of a user account.

| Value    | Meaning                                                                                            |
|----------|----------------------------------------------------------------------------------------------------|
| ACTIVE   | Can sign in.                                                                                       |
| LOCKED   | Temporarily locked after repeated failed sign-ins; unlocks at locked_until or by an administrator. |
| DISABLED | Cannot sign in until re-activated.                                                                 |

Used by: app\_user.status

### AuthSource: where the password is verified.

| Value     | Meaning                         |
|-----------|---------------------------------|
| LOCAL     | Argon2id hash held in app_user. |
| DIRECTORY | Active Directory / LDAP bind.   |

Used by: app\_user.auth\_source

### Audience: application a role or user belongs to.

| Value      | Meaning             |
|------------|---------------------|
| PORTAL     | Agent/Banca Portal. |
| BACKOFFICE | Back-office.        |

Used by: role.audience

**Channel: distribution channel of an agency.**

| Value  | Meaning                              |
|--------|--------------------------------------|
| AGENCY | Agency channel.                      |
| BANCA  | Bancatakaful channel (bank partner). |

Used by: agency.channel

**AgencyStatus: state of an agency or bank.**

| Value     | Meaning                                  |
|-----------|------------------------------------------|
| ACTIVE    | Trading; users can sign in.              |
| INACTIVE  | Not trading; users cannot sign in.       |
| SUSPENDED | Suspended by IIFT; users cannot sign in. |

Used by: agency.status

**AgentType: position in the distribution hierarchy.**

| Value      | Meaning                              |
|------------|--------------------------------------|
| MAIN_AGENT | Main agent; may have sub-agents.     |
| SUB_AGENT  | Sub-agent reporting to a main agent. |
| BANKER     | Bank officer.                        |

Used by: agent.agent\_type

**AgentStatus: registration status of an agent.**

| Value      | Meaning                         |
|------------|---------------------------------|
| PENDING    | Registration awaiting approval. |
| ACTIVE     | Approved and able to transact.  |
| INACTIVE   | Deactivated; cannot sign in.    |
| SUSPENDED  | Suspended; cannot sign in.      |
| TERMINATED | Contract ended.                 |
| REJECTED   | Registration rejected.          |

Used by: agent.status

**IdType: identification document type.**

| Value        | Meaning                                    |
|--------------|--------------------------------------------|
| NRIC         | Brunei identity card (IC).                 |
| PASSPORT     | Passport.                                  |
| BUSINESS_REG | Business registration number of a company. |

Used by: agent.id\_type, participant.id\_type

**ParticipantType: kind of participant.**

| Value      | Meaning                  |
|------------|--------------------------|
| INDIVIDUAL | Natural person.          |
| CORPORATE  | Company or organisation. |

Used by: participant.type

**AmlStatus: aML/KYC outcome held on agents and participants.**

| Value        | Meaning                                                        |
|--------------|----------------------------------------------------------------|
| NOT_SCREENED | Not yet screened.                                              |
| CLEAR        | No match, or match cleared by Compliance.                      |
| FLAGGED      | Potential match awaiting Compliance review; blocks submission. |
| REJECTED     | Match confirmed by Compliance; business is refused.            |

Used by: agent.aml\_status, participant.aml\_status

**AmlCaseStatus: state of a screening case.**

| Value           | Meaning                                               |
|-----------------|-------------------------------------------------------|
| AUTO_CLEARED    | Score below the threshold; no review needed.          |
| PENDING_REVIEW  | Score at or above the threshold; awaiting Compliance. |
| CLEARED         | Compliance decided the match is false.                |
| CONFIRMED_MATCH | Compliance confirmed the match.                       |

Used by: aml\_screening.status

**PolicyStatus: lifecycle status of a quotation or policy.**

| Value            | Meaning                                                                |
|------------------|------------------------------------------------------------------------|
| DRAFT            | Quotation being prepared; editable.                                    |
| PENDING_APPROVAL | Referred to IIFT for approval or quality check.                        |
| PENDING_PAYMENT  | Accepted; pay-first product awaiting verified payment before issuance. |
| ACTIVE           | Issued and in force.                                                   |
| REJECTED         | Rejected by a checker; can be reopened as a draft.                     |
| EXPIRED          | Cover period ended.                                                    |
| CANCELLED        | Cancelled after an approved cancellation request.                      |

Used by: policy.status, policy\_event.from\_status, policy\_event.to\_status

**PolicyPaymentStatus: payment state of a policy.**

| Value                | Meaning                               |
|----------------------|---------------------------------------|
| UNPAID               | No verified payment and none pending. |
| PENDING_VERIFICATION | Payment submitted, awaiting Finance.  |
| PAID                 | Fully paid.                           |

Used by: policy.payment\_status

**NomineeRole: role of a person named on a policy.**

| Value       | Meaning      |
|-------------|--------------|
| NOMINEE     | Nominee.     |
| BENEFICIARY | Beneficiary. |
| EXECUTOR    | Executor.    |

Used by: nominee.role

**PaymentMethod: how a payment was made.**

| Value         | Meaning                   |
|---------------|---------------------------|
| BANK_TRANSFER | Bank transfer.            |
| CHEQUE        | Cheque.                   |
| CASH_DEPOSIT  | Cash deposited at a bank. |
| ONLINE        | Online payment.           |

Used by: payment.method

### PaymentStatus: verification state of a payment.

| Value                | Meaning                        |
|----------------------|--------------------------------|
| PENDING_VERIFICATION | Awaiting Finance verification. |
| VERIFIED             | Verified; receipts issued.     |
| REJECTED             | Rejected with a reason.        |

Used by: payment.status

### ClaimStatus: state of a claim notification.

| Value        | Meaning                                     |
|--------------|---------------------------------------------|
| SUBMITTED    | Notified by the agent.                      |
| UNDER_REVIEW | Being reviewed by IIFT.                     |
| ACKNOWLEDGED | Accepted and registered in the core system. |
| REJECTED     | Not accepted.                               |
| CLOSED       | Closed.                                     |

Used by: claim.status

### DocumentOwnerType: type of record a document is attached to.

| Value       | Meaning                     |
|-------------|-----------------------------|
| AGENT       | Agent profile.              |
| AGENCY      | Agency or bank.             |
| PARTICIPANT | Participant.                |
| POLICY      | Quotation or policy.        |
| PAYMENT     | Payment.                    |
| CLAIM       | Claim notification.         |
| ISSUE       | Issue.                      |
| REPORT      | EOD run or report schedule. |

Used by: document.owner\_type

**DocumentStatus: verification state of a document.**

| Value    | Meaning                             |
|----------|-------------------------------------|
| UPLOADED | Uploaded, not yet checked.          |
| VERIFIED | Accepted by a checker.              |
| REJECTED | Rejected by a checker with remarks. |

Used by: document.status

**ApprovalType: transaction types that run through maker-checker.**

| Value                | Meaning                                                |
|----------------------|--------------------------------------------------------|
| AGENT_REGISTRATION   | New agent or bank officer.                             |
| AGENT_PROFILE_UPDATE | Change to agent details or reporting line.             |
| AGENT_STATUS_CHANGE  | Activation, suspension, deactivation or termination.   |
| PARTICIPANT_UPDATE   | Change to participant details.                         |
| POLICY_REFERRAL      | Referred application or quality check before issuance. |
| POLICY_ENDORSEMENT   | Policy endorsement.                                    |
| POLICY_CANCELLATION  | Policy cancellation with refund.                       |
| PAYMENT_VERIFICATION | Verification of a payment and receipt issuance.        |

Used by: workflow\_definition.type, approval\_request.type

**ApprovalStatus: state of an approval request.**

| Value     | Meaning                                           |
|-----------|---------------------------------------------------|
| PENDING   | Awaiting a decision at the current level.         |
| APPROVED  | All levels approved; the change has been applied. |
| REJECTED  | Rejected at a level, with remarks.                |
| WITHDRAWN | Withdrawn by the maker.                           |

Used by: approval\_request.status

**ApprovalActionType: action recorded on an approval request.**

| Value   | Meaning         |
|---------|-----------------|
| SUBMIT  | Request raised. |
| APPROVE | Level approved. |

| Value    | Meaning            |
|----------|--------------------|
| REJECT   | Request rejected.  |
| WITHDRAW | Request withdrawn. |

Used by: approval\_action.action

### IssuePriority: issue priority; selects SLA hours.

| Value    | Meaning                                        |
|----------|------------------------------------------------|
| CRITICAL | Service down or critical function unavailable. |
| HIGH     | Major function impaired.                       |
| MEDIUM   | Function impaired with a workaround.           |
| LOW      | Minor or cosmetic.                             |

Used by: issue.priority

### IssueStatus: state of an issue.

| Value       | Meaning                       |
|-------------|-------------------------------|
| OPEN        | Reported, not assigned.       |
| ASSIGNED    | Assigned to a person or team. |
| IN_PROGRESS | Being worked on.              |
| RESOLVED    | Resolution provided.          |
| CLOSED      | Confirmed and closed.         |

Used by: issue.status

### NotificationChannel: delivery channel.

| Value  | Meaning                                 |
|--------|-----------------------------------------|
| IN_APP | Bell icon in the portal or back-office. |
| EMAIL  | E-mail through the SMTP relay.          |
| SMS    | SMS through the SMS gateway.            |

Used by: notification.channel

**DeliveryStatus: delivery state of a notification.**

| Value   | Meaning                        |
|---------|--------------------------------|
| PENDING | Waiting for delivery or retry. |
| SENT    | Accepted by the gateway.       |
| FAILED  | Attempts exhausted.            |

Used by: notification.status

**IntegrationSystem: external system of an integration message or log entry.**

| Value     | Meaning                         |
|-----------|---------------------------------|
| CORE      | IITH core takaful system.       |
| FINANCE   | Financial system (FIN).         |
| AML       | External AML screening service. |
| EMAIL     | SMTP relay.                     |
| SMS       | SMS gateway.                    |
| DIRECTORY | Active Directory / LDAP.        |

Used by: outbox\_message.system, integration\_log.system, reconciliation\_run.system

**OutboxStatus: delivery state of an outbox message.**

| Value   | Meaning                                                                  |
|---------|--------------------------------------------------------------------------|
| PENDING | Waiting for first delivery.                                              |
| SENT    | Delivered.                                                               |
| FAILED  | Failed; retry scheduled at next_attempt_at.                              |
| DEAD    | Dead-letter: attempts exhausted or error not retryable; support alerted. |

Used by: outbox\_message.status

**IntegrationDirection: direction of an integration call.**

| Value    | Meaning                           |
|----------|-----------------------------------|
| OUTBOUND | Portal calls the external system. |
| INBOUND  | External system calls the portal. |

Used by: integration\_log.direction

**JobStatus: state of an end-of-day run.**

| Value     | Meaning                                |
|-----------|----------------------------------------|
| RUNNING   | In progress.                           |
| COMPLETED | Finished successfully.                 |
| FAILED    | Finished with an error; can be re-run. |

Used by: eod\_run.status

**ReconciliationStatus: outcome of a reconciliation run.**

| Value    | Meaning                                          |
|----------|--------------------------------------------------|
| MATCHED  | Every receipt was found in the financial system. |
| MISMATCH | One or more receipts were not found.             |

Used by: reconciliation\_run.status

**ParameterType: data type of a configuration parameter.**

| Value   | Meaning         |
|---------|-----------------|
| STRING  | Text.           |
| INTEGER | Whole number.   |
| DECIMAL | Decimal number. |
| BOOLEAN | true or false.  |

Used by: config\_parameter.value\_type

**ReportFrequency: how often a scheduled report runs.**

| Value   | Meaning                            |
|---------|------------------------------------|
| DAILY   | Every day at 07:00 Brunei time.    |
| WEEKLY  | Every Monday at 07:00.             |
| MONTHLY | On the 1st of each month at 07:00. |

Used by: report\_schedule.frequency

**ExportFormat: report output format.**

| Value | Meaning                 |
|-------|-------------------------|
| XLSX  | Excel workbook.         |
| CSV   | Comma-separated values. |
| PDF   | PDF document.           |

Used by: report\_schedule.format

**CommissionStatus: payment state of a commission accrual.**

| Value   | Meaning                            |
|---------|------------------------------------|
| ACCRUED | Accrued at issuance, not yet paid. |
| PAID    | Payment confirmed by Finance.      |

Used by: commission.status

## 6. Other Database Objects

### 6.1 Sequences

Business reference numbers come from PostgreSQL sequences, which are safe under concurrency across API replicas. A rolled-back transaction can leave a gap; numbers are unique but not guaranteed contiguous.

| Sequence           | Used for                   | Column                      | Format example                                 |
|--------------------|----------------------------|-----------------------------|------------------------------------------------|
| seq_agent_code     | Agent codes                | agent.agent_code            | AG-000045 (agency) or BK-000012 (bank officer) |
| seq_participant_no | Participant numbers        | participant.participant_no  | PT/26/000123                                   |
| seq_quotation_no   | Quotation numbers          | policy.quotation_no         | QT/26/000123                                   |
| seq_policy_no      | Policy numbers             | policy.policy_no            | PRO/26/000045 (product code prefix)            |
| seq_payment_no     | Payment references         | payment.payment_no          | PY/26/000123                                   |
| seq_receipt_no     | Receipt numbers            | receipt.receipt_no          | RC/26/000123                                   |
| seq_claim_no       | Claim notification numbers | claim.claim_no              | CL/26/000123                                   |
| seq_issue_no       | Issue numbers              | issue.issue_no              | IS/26/000123                                   |
| seq_request_no     | Approval request numbers   | approval_request.request_no | RQ/26/000123                                   |

Table 5: Sequences

### 6.2 Audit trail trigger

Function **audit\_log\_block\_mutation** (returns trigger) raises an exception; trigger **audit\_log\_no\_update** calls it BEFORE UPDATE OR DELETE on audit\_log, for each row. Any attempt to change or delete audit rows fails, whichever client issues it. Removing audit rows at the end of the retention period is done by a controlled archive procedure run by the database owner with the trigger disabled inside one audited maintenance transaction (delivered during implementation).

```
CREATE FUNCTION "audit_log_block_mutation"() RETURNS trigger AS $$
BEGIN
    RAISE EXCEPTION 'audit_log is append-only';
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER "audit_log_no_update"
BEFORE UPDATE OR DELETE ON "audit_log"
FOR EACH ROW EXECUTE FUNCTION "audit_log_block_mutation"();
```

## 6.3 Check constraints

| Constraint                         | Table              | Expression                                                               | Rule                                                                    |
|------------------------------------|--------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|
| policy_amounts_non_negative        | policy             | "contribution" >= 0 AND "sum_covered" >= 0 AND "outstanding_amount" >= 0 | Contribution, sum covered and outstanding amount can never be negative. |
| payment_amount_positive            | payment            | "total_amount" > 0                                                       | A payment must be for a positive amount.                                |
| payment_allocation_amount_positive | payment_allocation | "amount" > 0                                                             | Each allocation of a payment to a policy must be positive.              |
| nominee_share_range                | nominee            | "share_percent" > 0 AND "share_percent" <= 100                           | A nominee's share is greater than 0 and at most 100 per cent.           |

Table 6: Check constraints

## 6.4 Session table

user\_session has the structure expected by connect-pg-simple (sid, sess, expire). It is created by the migration, not by the library (createTableIfMissing is false), so the application database role needs no DDL rights. Expired sessions are pruned every 15 minutes.

## 7. Protected Data

Identification numbers are never stored in clear. The application encrypts them with AES-256-GCM using FIELD\_ENCRYPTION\_KEY before they reach the database, with a random 96-bit IV per value; the stored form is v1:<base64 of IV, authentication tag and ciphertext>, and the version prefix allows key rotation. To find a record by identification number and to enforce uniqueness, a blind index is stored alongside: an HMAC-SHA256 of the normalised number (spaces, hyphens and slashes removed, upper case) keyed with FIELD\_HASH\_KEY. The unique constraints (id\_type, id\_number\_hash) on agent and participant therefore prevent duplicates without decrypting. Screens show only the last four characters.

| Table             | Column         | Protection                                         |
|-------------------|----------------|----------------------------------------------------|
| agent             | id_number_enc  | Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)      |
| participant       | id_number_enc  | Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)      |
| nominee           | id_number_enc  | Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)      |
| agent             | id_number_hash | Blind index (HMAC-SHA256, FIELD_HASH_KEY)          |
| participant       | id_number_hash | Blind index (HMAC-SHA256, FIELD_HASH_KEY)          |
| app_user          | password_hash  | One-way hash (Argon2id)                            |
| password_history  | password_hash  | One-way hash (Argon2id)                            |
| signature_request | token_hash     | One-way hash (SHA-256 of one-time token)           |
| user_session      | sess           | Holds the CSRF token; never exposed to the browser |
| notification      | body           | Removed after delivery when sensitive = true       |

Table 7: Protected columns

Document files are encrypted separately with DOCUMENT\_ENCRYPTION\_KEY (AES-256-GCM, one IV per file); the document table holds only metadata and a generated storage key. The audit service writes **[REDACTED]** in place of passwords, hashes, encrypted identifiers, tokens and storage keys.